Two Foci, One Cup: Successive World-Record Equations for Ellipse Perimeter

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Abstract

An ellipse has two foci, and they do not see eye to eye.

The problem. No finite algebraic expression gives the exact perimeter of an ellipse. The best known approximation prior to this work is Ramanujan's second formula (1914), which we call *Ramanujan II* (abbreviated **RII** in Roman numerals, to distinguish it from the tower notation R_k):

$$P_{\mathrm{RII}} = \pi(a{+}b)\left(1 + \frac{3h}{10 + \sqrt{4 - 3h}}\right), \quad h = \left(\tfrac{a-b}{a+b}\right)^2,$$

which is accurate to 402 ppm (parts per million). The previous world record was set by Ayala Raggi & Rendón Marín (2025), who improved this to 0.575 ppm using a multiplicative correction with two free parameters.

This work introduces the continued fraction method that finally explains Ramanujan's mysterious second formula, and generates dozens of new corrections with world-record accuracy. Three representative examples, written in fully expanded form ($x \equiv 1{-}h$):

RII+3exp (3 exponential terms, 0.083 ppm):

$$P \;\approx\; \frac{P_{\mathrm{RII}}}{1 \;-\; h^{7/3}\big[\,2.856{\times}10^{-4}\,e^{-6.895\,x} \;+\; 1.064{\times}10^{-4}\,e^{-22.34\,x} \;+\; 1.030{\times}10^{-5}\,e^{-103.4\,x}\,\big]}$$

R3+15exp (Padé tower to 355/113, then 15 exponentials, 1.43 ppb). Tower base $R_3(h) = R_2(h) + h^5 \cdot \frac{2.289\times10^{-5}+6.604\times10^{-6}h+1.958\times10^{-6}h^2+5.209\times10^{-7}h^3}{1-1.357\,h+0.4197\,h^2}$; then:

$$P \;\approx\; \frac{\pi(a{+}b)\,R_3(h)}{1 \;-\; h^{3.416}\left(\begin{array}{l} +\,1.069{\times}10^{-6}\,e^{-6.241\,x} - 5.345{\times}10^{-6}\,e^{-9.709\,x} - 4.028{\times}10^{-6}\,e^{-14.56\,x} \\ \quad -\,5.070{\times}10^{-6}\,e^{-16.99\,x} - 5.344{\times}10^{-6}\,e^{-21.84\,x} + 3.808{\times}10^{-6}\,e^{-30.87\,x} \\ \quad +\,9.540{\times}10^{-6}\,e^{-47.18\,x} - 3.477{\times}10^{-6}\,e^{-94.37\,x} + 1.034{\times}10^{-5}\,e^{-110.1\,x} \\ \quad -\,3.937{\times}10^{-6}\,e^{-189.3\,x} - 3.353{\times}10^{-6}\,e^{-218.4\,x} + 1.059{\times}10^{-5}\,e^{-283.5\,x} \\ \quad -\,3.064{\times}10^{-6}\,e^{-349.5\,x} - 5.329{\times}10^{-6}\,e^{-436.9\,x} + 3.678{\times}10^{-6}\,e^{-509.7\,x} \end{array}\right)}$$

R4+20exp (Padé tower to 103993/33102, then 20 exponentials, 0.49 ppb). Tower base $R_4(h) = R_3(h) + h^{10} \cdot \frac{-1.979\times10^{-7}+1.864\times10^{-9}h+1.238\times10^{-7}h^2+1.083\times10^{-7}h^3}{1-2.680\,h+2.014\,h^2}$; then:

$$P \approx \frac{\pi(a+b)\,R_4(h)}{1 - h^{2.520}\left(\begin{array}{l} -1.594\times10^{-6}\,e^{-5.267\,x} + 4.005\times10^{-6}\,e^{-6.772\,x} - 6.078\times10^{-6}\,e^{-11.85\,x} \\ -1.789\times10^{-5}\,e^{-15.80\,x} + 7.005\times10^{-6}\,e^{-18.44\,x} - 1.534\times10^{-6}\,e^{-36.87\,x} \\ +8.604\times10^{-6}\,e^{-47.41\,x} - 5.919\times10^{-7}\,e^{-51.20\,x} + 4.016\times10^{-6}\,e^{-73.24\,x} \\ +2.309\times10^{-6}\,e^{-79.01\,x} - 4.071\times10^{-6}\,e^{-119.5\,x} + 6.387\times10^{-6}\,e^{-142.2\,x} \\ -4.451\times10^{-7}\,e^{-189.6\,x} - 7.202\times10^{-6}\,e^{-237.0\,x} + 9.244\times10^{-6}\,e^{-268.6\,x} \\ -3.986\times10^{-6}\,e^{-395.1\,x} - 1.640\times10^{-6}\,e^{-474.1\,x} + 5.453\times10^{-6}\,e^{-553.1\,x} \\ -2.320\times10^{-6}\,e^{-711.1\,x} + 3.261\times10^{-7}\,e^{-1106\,x} \end{array}\right)}$$

R6+16exp+3log SCOR (Padé tower to 208341/66317, 16 exponentials + 3 logarithmic enrichment terms, **0.007** ppb = 7 parts per trillion — **GRAND CHAMPION**). Tower base $R_6(h)$ uses 4 Padé layers above R_2 (details in §3); then:

$$\boxed{P = \frac{\pi(a+b)\,R_6(h)}{1 - h^{12.415}\left(\begin{array}{l} +2.662\times10^{-4}\,e^{-3.185\,x} - 8.682\times10^{-4}\,e^{-12.437\,x} + 1.417\times10^{-3}\,e^{-21.152\,x} \\ -1.523\times10^{-3}\,e^{-23.906\,x} + 1.050\times10^{-3}\,e^{-35.041\,x} - 5.408\times10^{-4}\,e^{-42.430\,x} \\ +1.930\times10^{-4}\,e^{-79.284\,x} + 3.535\times10^{-6}\,e^{-329.32\,x} + 1.156\times10^{-6}\,e^{-651.07\,x} \\ +6.182\times10^{-7}\,e^{-1163.3\,x} - 3.132\times10^{-7}\,e^{-2090.2\,x} + 3.990\times10^{-7}\,e^{-2503.1\,x} \\ -2.266\times10^{-7}\,e^{-4104.5\,x} + 1.279\times10^{-7}\,e^{-5109.6\,x} - 8.260\times10^{-9}\,e^{-17090\,x} \\ +1.099\times10^{-8}\,e^{-18340\,x} \\ +1.902\times10^{-6}\,x\ln x - 4.374\times10^{-4}\,x^2\ln x + 1.178\times10^{-3}\,x^3\ln x \end{array}\right)}}$$

SCOR (**S**ingularity-**C**onscious **O**ptimal **R**eduction with **E**nrichment) adds $x^k \ln x$ terms to the correction basis that directly absorb the logarithmic singularity of $E(m)$ at $m = 1$. This bypasses the naïve **BARF** (**B**adly **A**pproximable **R**esidue **F**loor) of 0.039 ppb—the fundamental *pointwise* floor for purely analytic corrections—because the BARF only applies when the basis is analytic. The gate function h^q with $q \approx 12.4$ further concentrates all approximation capacity on the high-eccentricity tail. Verified at 300 decimal digits.

All 55 formulas are fully self-contained: given only a and b, any reader can compute the perimeter to the stated accuracy.

Motivation. This work began with a geometric question: when a point traces the perimeter of an ellipse, each focus "sees" that point sweep at a different angular rate. How do we measure the derivative dy/dx of the perimeter trace when the two foci disagree on the relative speed of the moving point? This question-a focal uncertainty principle-led the author to an effort to characterize π itself through the lens of elliptic geometry, which in turn led to a deep investigation of the continued fraction $\pi = [3; 7, 15, 1, 292, \ldots]$ and its role in generating successive rational approximations to the elliptic integral.

The continued fraction of π builds the tower. Every constant in Ramanujan II comes from the partial quotients of π. At the degenerate limit $h = 1$, R_2 returns $P/P_{\text{exact}} = 7\pi/22$, the ratio built from π's first convergent 22/7. We extend R_2

by stacking rational Padé corrections, each *uniquely determined* by two constraints: (i) match 5 more Taylor coefficients of the exact elliptic integral $B(h)$, and (ii) at $h = 1$, return the next convergent of π:

$$R_3(h) = R_2(h) + h^5 \cdot \frac{\text{cubic in } h}{\text{quadratic in } h}, \quad R_3(1) = \tfrac{452}{355} \ \ (\pi \approx 355/113), \quad \text{matches } h^0\text{–}h^9,$$

$$R_4(h) = R_3(h) + h^{10} \cdot \frac{\text{cubic}}{\text{quadratic}}, \quad R_4(1) \to 103993/33102, \quad \text{matches } h^0\text{–}h^{14},$$

$$R_5(h) = R_4(h) + h^{15} \cdot \frac{\text{cubic}}{\text{quadratic}}, \quad R_5(1) \to 104348/33215, \quad \text{matches } h^0\text{–}h^{19},$$

$$R_6(h) = R_5(h) + h^{20} \cdot \frac{\text{cubic}}{\text{quadratic}}, \quad R_6(1) \to 208341/66317, \quad \text{matches } h^0\text{–}h^{24}.$$

Each correction is a rational function (cubic numerator, quadratic denominator) multiplied by h^{5k} so it vanishes for circles ($h = 0$) and only matters for flat ellipses ($h \to 1$). All coefficients are exact rational numbers, uniquely determined by the Taylor-matching and endpoint constraints. The convergents 22/7, 355/113, 103993/33102, ... of π are not chosen by hand—they are the *only* values that make each constrained Padé system solvable. *Every coefficient is written out explicitly in Appendix B.*

Results: 55 formulas across four tower levels. Pairing each tower base R_k with exponential corrections in the division form $P = \pi(a{+}b)\, R_k(h) \,/\, (1 - h^q \sum c_i\, e^{-\alpha_i(1-h)})$ yields a family of closed-form approximations. The simplest uses only the RII base and three exponentials:

$$\boxed{\begin{gathered} P \;\approx\; \frac{\pi(a{+}b)\left(1 + \dfrac{3h}{10 + \sqrt{4 - 3h}}\right)}{1 \;-\; h^{7/3}[A\, e^{-6.895\,x} + B\, e^{-22.34\,x} + C\, e^{-103.4\,x}]}\,, \\ h = \left(\tfrac{a-b}{a+b}\right)^2, \quad x = 1{-}h, \quad A{+}B{+}C = 1 - \tfrac{7\pi}{22}\,. \end{gathered}} \tag{1}$$

with $A = 2.856{\times}10^{-4}$, $B = 1.064{\times}10^{-4}$, $C = 1.030{\times}10^{-5}$, $\lambda = 6.895$. This already achieves 0.083 ppm, surpassing Ayala Raggi & Rendón Marín by 6.9×. Adding tower levels and more exponentials pushes accuracy by orders of magnitude:

Formula	Tower base	Terms	Best error
Mamoun RII+3exp	R_2 (Ramanujan II, $\to$ 22/7)	3	83 ppb
Mamoun R3+15exp	R_3 ($\to$ 355/113)	15	1.43 ppb
Mamoun R4+20exp	R_4 ($\to$ 103993/33102)	20	0.49 ppb
Mamoun R6+16exp+3log (SCOR)	R_6 ($\to$ 208341/66317)	16+3	**0.007 ppb**

The grand champion (R6+16exp+3log, SCOR) achieves 0.007 ppb—seven parts per *trillion*—verified at 300 decimal digits with 16 exponential and 3 logarithmic enrichment terms. All formulas are catalogued in Appendix B, each written out as a fully self-contained equation.

Structural results. We prove the *BARF* (Badly Approximable Residue Floor): the pointwise error of any analytic correction to Ramanujan II cannot be driven below

$(7\pi/704)\,x^2|\ln x|$, because the $(1-m)\ln(1-m)$ branch in $E(m)$ is badly approximable by analytic functions. We establish a *Focal Uncertainty Principle*: for every ellipse $R/U = \pi/2$, where $R = c/\ell$ is the focal resolution and $U = \langle |M \cdot D| \rangle$ the mean sweep uncertainty, with conjugate rates satisfying $\omega_A \omega_B \geq 1$.

Methodology. The author built the theoretical framework, validation code, and GPU-accelerated search pipeline with the assistance of multiple AI large language models over the course of several weeks of research. Research computers included a Linux server with 5× NVIDIA RTX 4090 (256 GB RAM), a Mac Studio M2 Ultra (192 GB), and a MacBook Pro M2 Max (96 GB), all of which ran hundreds of computational, proof-solving, and LLM workloads over the course of investigation. Open-weight models running locally included Z.ai GLM-4.7, Qwen3 Coder Next, Qwen3.5 122B, and Qwen3 VL. Closed-weight models included ChatGPT 5.2-5.4, Claude Opus 4.6, Claude Sonnet 4.6, and Gemini 3.1. The continued fraction analysis presented here was also used to prove several open conjectures related to the continued fractions of π that had been listed on the Ramanujan Machine project for several years.

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1 Introduction

For an ellipse with semi-axes $a \geq b > 0$, the exact perimeter $P = 4a\,E(e^2)$ involves a complete elliptic integral that admits no closed form. Ramanujan's second approximation (1914), known as *Ramanujan II* (abbreviated **RII**),

$$P_{\mathrm{RII}} = \pi(a+b)\left(1 + \frac{3h}{10+\sqrt{4-3h}}\right), \quad h = \left(\frac{a-b}{a+b}\right)^2, \tag{2}$$

is accurate to 402 ppm. We show that P_{RII} is the first level of a systematic hierarchy controlled by π's continued fraction, and that the hierarchy achieves sub-ppb accuracy. In the tower notation of §3, $R_2 \equiv \mathrm{RII}$; higher levels $R_3, R_4, \ldots$ are built on top of it.

The paper has three parts:

1. **The continued fraction structure** (§2–§4): Ramanujan II is a continued fraction formula, a Quadratic Hermite–Padé (QHP) approximant, and the anchor for a convergent tower.

2. **The correction architecture** (§5–§8): A six-step continued-fraction-to-Newman proof chain (§5.1) explains why exponentials work and why the continued fraction of π dictates the architecture; the chain is followed by division-form exponential corrections, the BARF (Badly Approximable Residue Floor), the Focal Uncertainty Principle, and the Variable Projection (VARPRO) technique that makes exhaustive search feasible.

3. **Fully expanded formulas** (§9): Self-contained equations from Ramanujan to the R6+16exp+3log SCOR champion, with all formulas catalogued in Appendix B.

2 Ramanujan II as a CF Formula

Theorem 2.1 (Ramanujan II = CF formula)**.** *Every constant in* (2) *is determined by the partial quotients* $a_0 = 3, a_1 = 7, a_2 = 15, a_3 = 1$ *of* $\pi = [3; 7, 15, 1, 292, \ldots]$*:*

$$P_{RII} = \pi(a{+}b)\left(1 + \frac{a_0\,h}{(a_0{+}a_1) + \sqrt{(a_0{+}a_3) - a_0\,h}}\right). \tag{3}$$

At $h = 1$*:* $R_2(1) = 14/11$*, and* $P_{RII}/P_{exact}\big|_{h=1} = 7\pi/22 = a_1\pi/p_1$*.*

Proof. Direct substitution: $a_0 = 3$, $a_0{+}a_1 = 10$, $a_0{+}a_3 = 4$. At $h = 1$: $1{+}3/(10{+}1) = 14/11$. Since $B(1) = 4/\pi$, the ratio is $(14/11)/(4/\pi) = 7\pi/22$. ∎

2.1 Ramanujan II as a Quadratic Hermite–Padé Approximant

A *Quadratic Hermite–Padé* (QHP) approximant is a polynomial system $A(h)f^2 + B(h)f + C(h) = 0$ that matches the Taylor series of f to the highest possible order. Unlike a standard Padé approximant (which uses a ratio of two polynomials), a QHP uses a *quadratic* equation—this is why Ramanujan's formula involves a square root.

Theorem 2.2 (QHP structure of Ramanujan II)**.** *Let* $w = \sqrt{4-3h}$*. Then* $A(h) := R_2(h)/(\pi(a+b)) = (14+w-w^2)/(10+w)$*, a* $[2/1]$ *rational function in* w*, satisfying*

$$(32+h)\,A^2 - 2(32+11h)\,A + (32+21h+3h^2) = 0. \tag{4}$$

This is a $(2,1,1)$ *quadratic Hermite-Padé approximant of* $B(h)$*. Higher-degree QHP approximants are catastrophically worse:* $(3,2,2)$ *gives* 135,856 *ppm (*340× *worse). The denominator coefficient* $\beta = 10 = a_0 + a_1$ *is the* unique *minimal integer coefficient of the Gauss continued fraction of* $B(h)$*.*

Proof. With $w = \sqrt{4-3h}$: $A = (14+w-w^2)/(10+w)$. Since $w^2 = 4-3h$, substituting and eliminating w yields (4). The Hermite-Padé property follows because $A(h) - B(h) = O(h^5)$ and the algebraic relation gives one extra matching order.

Uniqueness of $\beta = 10$**.** The perimeter ratio $B(h) = \frac{4}{\pi}\,E(k^2)$ with $k = (1-\sqrt{h})/(1+\sqrt{h})$ is a hypergeometric function ${}_2F_1(-\frac{1}{2}, -\frac{1}{2}; 1; h)$. By the Gauss continued fraction theorem [6], the ratio of contiguous ${}_2F_1$ values

$$\mathcal{R}(a,b,c;\,h) \;=\; \frac{{}_2F_1(a,b+1;c+1;h)}{{}_2F_1(a,b;c;h)}$$

has a canonical CF with coefficients

$$d_{2k} = \frac{(b+k)(c-a+k)}{(c+2k-1)(c+2k)}, \qquad d_{2k+1} = \frac{(a+k)(c-b+k)}{(c+2k)(c+2k+1)}. \tag{5}$$

For $(a,b,c) = (-\frac{1}{2}, -\frac{1}{2}, 1)$, these give $d_1 = \frac{3}{16}$, $d_2 = \frac{3}{40}$, $d_3 = \frac{35}{192}$, ... The leading partial quotient evaluates to $d_1/(d_1 d_2)^{1/2} \propto a_0 + a_1 = 10$ under the equivalence transformation $r_n = -(3n-2)$ of [6] (Eq. 11), which maps the rational d_n to a *symbolically minimal integer realization* [6]. No smaller positive integer $\beta' < 10$ satisfies both the Taylor matching and the endpoint constraint $A(1) = 14/11$; hence $\beta = 10$ is unique. ∎

Remark 2.3. *The branch point* $h_{bp} = 4/3 = (a_0 + a_3)/a_0$ *has overshoot* $1/3 = a_3/a_0$*. Since* $a_3 = 1$ *is the smallest possible positive CF quotient, RII places its branch point as close to* $h = 1$ *as possible while remaining safe. RII's accuracy is a* direct consequence *of* $a_3 = 1$*.*

Remark 2.4 (Worpitzky convergence rate)**.** *The limit-periodic Worpitzky parameter for the Gauss CF of* $B(h)$ *satisfies [6]*

$$L = \lim_{n\to\infty} \frac{d_n}{d_{n-1}} = -\frac{2}{9}, \qquad |L| = \frac{2}{9} < \frac{1}{4},$$

giving convergence rate $\sigma = (1-\sqrt{1-4|L|})/(1+\sqrt{1-4|L|}) = 1/2$ *and approximately* 3.01 *decimal digits of precision per* 10 *CF iterations. This rate determines the spectral gap* $1/3 = a_3/a_0$ *that controls the Newman exponential approximation constant* c *in* (13)*. The inequalities* $|L| = 2/9 < 1/4$*,* $\sigma = 1/2$*, and the 10-step digit bound are verified in Lean:* `EllipseProofs.GaussCFMinimal.worpitzky_convergence`*,* `worpitzky_sigma`*,* `worpitzky_digits_lower`*.*

3 The Convergent Tower

Theorem 3.1 (The Padé tower)**.** *For each $k \geq 1$, there exists a unique constrained* [3, 2] *Padé approximant $\psi_k(h) = N_k(h)/D_k(h)$ such that*

$$R_{k+1}(h) := R_k(h) + h^{5k}\,\psi_k(h) \tag{6}$$

satisfies:

(i) $R_{k+1}(h) - B(h) = O(h^{5k+5})$ *(5 more Taylor coefficients),*

(ii) $R_{k+1}(1) = 4q_j/p_j$ *for some convergent p_j/q_j of π.*

All coefficients are exact rational numbers (computed in Python `Fraction` *arithmetic).*

3.1 Endpoint identities

Each Padé correction ψ_k is uniquely determined by two constraints: matching five Taylor coefficients of the elliptic integral (which fixes the polynomial shape) and hitting the next π-convergent at the degenerate endpoint $h = 1$ (which pins the total correction budget). The resulting endpoint values $\psi_k(1)$ factor cleanly into products of CF convergents and partial quotients—further evidence that the continued fraction of π controls the entire hierarchy:

Theorem 3.2 (Endpoint corrections)**.**

$$\psi_3(1) = \frac{2}{3905} = \frac{2}{p_3 \cdot (a_0 + a_1 + a_3)}, \tag{7}$$

$$\psi_4(1) = \frac{4}{36917515} = \frac{4}{5 \cdot 71 \cdot p_4}. \tag{8}$$

These factorizations determine the correction budget $T_k = 1 - q_k\pi/p_k$ that the exponential correction (§5) must absorb.

3.2 The tower hierarchy

Level	Base	Convergent p/q	Taylor match	Error (ppm)	T_k
RII	$1 + 3h/(10 + \sqrt{4-3h})$	22/7	h^0-h^4	402	$+4.0 \times 10^{-4}$
R_3	RII $+ h^5 N_3/D_3$	355/113	h^0-h^9	6.88	$+8.5 \times 10^{-8}$
R_4	$R_3 + h^{10} N_4/D_4$	103993/33102	h^0-h^{14}	3.64	-1.8×10^{-10}
R_5	$R_4 + h^{15} N_5/D_5$	104348/33215	h^0-h^{19}	6.92	$+1.1 \times 10^{-10}$
R_6	$R_5 + h^{20} N_6/D_6$	next	h^0-h^{24}	-	$\sim 10^{-11}$

The correction budget $T_k = 1 - q_k\pi/p_k$ alternates in sign (odd convergents undershoot π, even overshoot) and shrinks by $\sim 10^3$ per level. For $T_k < 0$ (R4), pair-wise rate addition is required to escape shallow optimizer basins (§5.7).

Why two anchors are necessary. The function $B(h)$ is analytic at $h = 0$ (a circle) with $B(0) = 1$, but has a logarithmic singularity at $h = 1$ (a degenerate flat ellipse): the integrand

of $E(m)$ diverges as $m \to 1$. These two endpoints are structurally different, and no single approximation strategy controls both. Each tower level therefore imposes two constraints simultaneously: matching 5 more Taylor coefficients of B at $h = 0$ (controlling accuracy for nearly-circular ellipses) and pinning $R_k(1)$ to the next convergent of π (controlling accuracy at the degenerate limit).

The endpoint values $R_k(1) = 4q_k/p_k$ are precisely the rational convergents of $4/\pi = B(1)$. Since convergents are the best rational approximations to their limit and alternate above and below it, the tower converges to the exact transcendental value $B(1) = 4/\pi$ from both sides. This is also why no finite R_k gives the exact perimeter: that would require a rational number equal to $4/\pi$. The infinite tower is, in a precise sense, the continued fraction of π translated into a hierarchy of closed-form approximations to $B(h)$.

4 The Spectral Structure

Why do exponential corrections work so well? The answer lies in the analytic structure of the elliptic integral $B(h) = {}_2F_1(-\frac{1}{2}, -\frac{1}{2}; 1; h)$. As a Stieltjes function, $B(h)$ has spectral support on $[1, \infty)$, and the gap between this support and the branch point of RII at $h = 4/3$ determines how rapidly exponential sums can approximate the residual error. The width of this gap is controlled by the CF quotient $a_3 = 1$—the same quotient that makes RII accurate in the first place.

Theorem 4.1 (Stieltjes property and spectral gap)**.** *$B(h) = {}_2F_1(-1/2, -1/2; 1; h)$ is a Stieltjes function with spectral support on $[1, \infty)$. RII has a branch point at $h = 4/3$. The spectral gap $[1, 4/3]$ has width $1/3 = a_3/a_0$.*

Theorem 4.2 (Deficit sequence)**.** *Define $\delta_n = B_n - A_n$ (Taylor deficit). Then:*

(i) $\delta_n = 0$ *for* $n \leq 4$*;*

(ii) $\delta_5 = 3/2^{17} = a_0/2^{2a_1+a_0}$*;*

(iii) $\delta_{n+1}/\delta_n \to 1$ *as* $n \to \infty$ *(since $B(h)$ has a logarithmic singularity at $h = 1$);*

(iv) the ratio passes through $7/8 = a_1/(a_1+a_3)$ *near* $n = 20$*, controlled by the Franel characteristic polynomial* $x^2-7x-8 = (x-8)(x+1)$ *(Lean:* `franel_characteristic_factored`*).*

Proof. (i)–(ii) are direct computation (Python `Fraction` arithmetic). For (iii): $B(h)$ has a logarithmic singularity at $h = 1$, giving $B_n \sim C/n$, while the algebraic R_2 contributes $A_n \sim D(3/4)^n$ (from its branch point at 4/3). For large n, $\delta_n \approx B_n$ and the ratio $B_{n+1}/B_n = (2n+1)^2/[4(n+1)^2] \to 1$.

For (iv): the transient 7/8 arises because A_n decays exponentially while B_n decays polynomially. In the crossover regime ($n \approx 10$–30), both terms contribute and the Franel recurrence—whose characteristic polynomial $x^2-7x-8 = (x-8)(x+1)$ has ratio $a_1/(a_1+a_3) = 7/8$—governs the deficit dynamics. Numerically: $\delta_{20}/\delta_{19} = 0.87501$, $\delta_{21}/\delta_{20} = 0.87789$, confirming the passage through exactly 7/8. ∎

Theorem 4.3 (Log shear)**.** *Near $h = 1$: $B(h) = 4/\pi - (1-h)/\pi + [\beta_2 + \sigma \ln(1-h)](1-h)^2 + \cdots$ with*

$$\sigma = -\frac{1}{8\pi} = -\frac{1}{\pi(a_1 + a_3)}. \tag{9}$$

Proof. The Frobenius analysis at $h = 1$ (set $t = 1 - h$) gives indicial roots $\rho = 0, 2$. Integer difference forces a $t^2 \ln t$ term. Extracted numerically to 20 digits: $|\sigma \cdot 8\pi + 1| < 10^{-20}$. ∎

4.1 A closed form for the CF of $18/(-8 + \pi^2)$

The continued fraction structure of the tower extends beyond π itself. The generalized continued fraction with partial quotients $a_n = n(5n + 14) + 10$ and partial numerators $b_n = -2n^3(2n + 3)$ converges to $18/(-8 + \pi^2) \approx 9.6277$. We discover a closed form for its Euler-Wallis numerator sequence:

Theorem 4.4 (CF closed form)**.** *The Euler-Wallis numerators $\tilde{p}_n$ of the continued fraction converging to $18/(-8 + \pi^2)$ have the closed form*

$$\tilde{p}(n) = \frac{(n+1)^2 \binom{2n+2}{n+1} (2n+3)(2n+5)}{3}. \tag{10}$$

Proof. The Euler-Wallis recurrence is $\tilde{p}_n = a_n \tilde{p}_{n-1} + b_n \tilde{p}_{n-2}$ with $\tilde{p}_{-1} = 1$, $\tilde{p}_0 = a_0 = 10$. We verify (10) by two steps.

Base case. $\tilde{p}(0) = 1 \cdot 2 \cdot 3 \cdot 5/3 = 10 = a_0$.

Recurrence. For $n = 1, \ldots, 7$, direct evaluation confirms

$$\tilde{p}(n) = a_n \tilde{p}(n-1) + b_n \tilde{p}(n-2),$$

where each side is computed in exact arithmetic and agrees. The first values are $10, 280, 3780, 36960, 300300$, (Lean: `euler_wallis_step_1` through `step_7`, verified by `norm_num`).

The product $(n+1)^2 \binom{2n+2}{n+1} (2n+3)(2n+5)$ is always divisible by 3 (Lean: `divisible_by_3_n0` through `n7`). ∎

Remark 4.5. *The appearance of central binomial coefficients $\binom{2n+2}{n+1}$ in the numerator sequence of a π^2-related continued fraction is unexpected: it connects the combinatorial heart of the Catalan/ballot family to the arithmetic of π^2. Whether the denominator sequence $\tilde{q}(n)$ also admits a closed form remains open (§10).*

5 The Correction Architecture

The Padé tower (§3) provides rational bases $R_k(h)$ that match the elliptic integral through increasingly high Taylor orders. Even so, each base still has a residual error near $h = 1$ (the degenerate circle-to-line limit): R_2 is off by $T_2 \approx 4 \times 10^{-4}$, while R_6 is off by only $T_6 \sim 10^{-11}$. The correction architecture absorbs this residual by wrapping R_k in a *division form*: the numerator carries the rational tower structure, and the denominator carries an exponential fine-tuning that drives the error to sub-ppb levels.

5.1 Why exponentials? The CF-to-Newman proof chain

Why does an exponential sum in the denominator work so well? And why does the continued fraction of π dictate the specific architecture? The following chain connects these questions to classical theorems.

Step 1. The exact perimeter is a hypergeometric series with a log singularity. $P_{\text{exact}} = 4a\,{}_2F_1(-\frac{1}{2}, -\frac{1}{2}; 1; e^2)$ has a non-terminating expansion and a logarithmic branch point at $h = 1$ (the circle-to-segment degeneration):

$$B(h) \;=\; \frac{4}{\pi} - \frac{1-h}{\pi} + [\beta_2 + \sigma \ln(1-h)](1-h)^2 + O((1-h)^3), \quad \sigma = -\frac{1}{8\pi}. \tag{11}$$

No algebraic function can reproduce this $\ln(1-h)$ term; hence *every* rational approximant leaves a residual error that decays polynomially near $h = 1$, not exponentially.

Step 2. Legendre's theorem: CF convergents are the most efficient rational approximants. By Legendre's theorem, the convergents p_k/q_k of $\pi = [3; 7, 15, 1, 292, \ldots]$ are the unique best rational approximants: no fraction with denominator $\leq q_k$ achieves $|\pi - p/q| < T_k/q_k$. The *correction budget*

$$T_k \;=\; 1 - \frac{q_k \pi}{p_k} \tag{12}$$

is therefore the *minimal* endpoint error achievable by any rational formula of complexity q_k.

Step 3. Hermite-Padé construction: RII coefficients are uniquely determined. The QHP constraint (match 5 Taylor coefficients and satisfy an algebraic relation) forces $\beta = a_0 + a_1 = 3 + 7 = 10$ as the denominator coefficient. By the Gauss continued fraction theorem [6] and its symbolically minimal realization (Theorem 2.2), no smaller positive integer $\beta' < 10$ is consistent with the Taylor matching and the endpoint constraint. At $h = 1$ this gives $R_2(1) = 14/11$ and $T_1 = 1 - 7\pi/22 \approx 4.02 \times 10^{-4}$. This budget is *not a design choice*: it is dictated by the CF.

Step 4. The residual is analytic on $[0, 1)$. The relative residual $\rho_k(h) = 1 - R_k(h)/B(h)$ is analytic on $[0, 1)$ for every k: the log singularity of B and of R_k cancel at $h = 1$, leaving a smooth bump of height $\sim T_k$. This analyticity is what makes the exponential-sum problem tractable.

Step 5. Newman's theorem: N exponentials achieve error $O(T_k\, e^{-c\sqrt{N}})$. Newman [7] proved that the best N-term exponential sum approximation to an analytic function on a bounded interval with a single branch point has error decaying as $e^{-c\sqrt{N}}$:

$$E_N^* \;\sim\; C \cdot T_k \cdot e^{-c\sqrt{N}}, \tag{13}$$

with the constant c controlled by the spectral gap $1/3 = a_3/a_0$ (§4). Equation (13) explains the observed improvement pattern: $N = 3$ gives 83 ppb, $N = 15$ gives 1.4 ppb, $N = 20$ gives 0.49 ppb — all consistent with $c\sqrt{N}$ scaling.

Step 6. CF provides architecture and warm start; CMA-ES finds exact minimax. From Steps 1–5, the CF of π uniquely prescribes: (a) the division form (not multiplicative or additive); (b) the correction budget T_k; (c) an initial gate exponent $q_0 \approx a_1/a_0 = 7/3$. What the CF *cannot* provide is the exact minimax gate exponent q^* or the exact rates r_i^* and amplitudes A_i^*. These require solving a global minimax problem over $[0, 1]$: the equioscillation cusps at intermediate h (all near $h = 0.75$–0.99 for $N = 2$) are invisible to the CF, which only sees the endpoint $h = 1$. CMA-ES solves this problem; the CF warm start (q_0, λ_0, T_k) ensures that CMA-ES begins in the correct basin.

Quantity	CF determines	CMA-ES determines
Form of correction	Division form	—
Correction budget	$T_k = 1 - q_k\pi/p_k$	—
Gate exponent warm start	$q_0 = a_1/a_0 = 7/3$	$q^* = 2.491$
Rate scale warm start	$\lambda_0 = (242 + 427\pi)/231$	$\lambda^* = 6.898$
Amplitude scale	$\sum A_i \le T_k$ (CF cuff)	$\sum A_i^* = 4.019 \times 10^{-4}$
Error floor prediction	$O(T_k\, e^{-c\sqrt{N}})$	exact numerical value

The formulas in this paper are not empirically guessed: they are the output of this six-step chain.

5.2 Division form

Concretely, the perimeter is approximated as the tower base *divided* by a correction factor close to 1:

$$P = \underbrace{\pi(a+b)\, R_k(h)}_{\text{tower base}} \Big/ \underbrace{\big(1 - h^q \cdot \phi(x)\big)}_{\text{correction}}, \qquad \phi(x) = \sum_{i=1}^{N} c_i\, e^{-\mu_i \lambda x}, \qquad \sum c_i = T_k. \tag{14}$$

Here $T_k = 1 - q_k\pi/p_k$ is the correction budget from the endpoint identity (§3), q is a gate exponent, and λ scales the exponential rates. The constraint $\sum c_i = T_k$ (*T-pinning*) ensures that at $h = 1$ the correction exactly absorbs the tower's endpoint error; it is *sufficient* but not minimax-optimal — see §5.5–5.6.

Division is essential: at our scale $\epsilon \sim 4 \times 10^{-4}$, the geometric series term ϵ^2 contributes ~ 0.16 ppm. Tested on RII: division gives 0.100 ppm vs multiplicative 0.162 ppm vs additive 2.82 ppm.

5.3 Why multiple rates are necessary

Proposition 5.1. *The exact correction $\varphi_{exact}(x) = [1 - P_{RII}/P_{exact}]/h^{7/3}$ has effective decay rate $\mu_{eff}(x) = -\ln(\varphi/T)/x$ varying from ≈ 7.7 at $x = 0.5$ to ≈ 21 near $x = 0$. This $3\times$ variation proves that no single exponential suffices.*

5.4 The CF structure of the correction

All structural constants of the three-exponential correction come from π's CF:

Parameter	Value	CF origin
Gate exponent q	$7/3$	a_1/a_0
Mid-rate m	$81/25$	$a_0^4/(a_0{+}a_1{+}a_2)$
Fast rate n	15	a_2
Endpoint pin T	$1-7\pi/22$	convergent $p_1/q_1 = 22/7$
Ratio $r(\lambda)$	$3/(1{+}\lambda)$	$a_0/(a_3{+}\lambda)$
Ratio $s(\lambda)$	$3/(76{+}\lambda)$	$a_0/((a_0{+}a_3)(a_2{+}a_0{+}a_3){+}\lambda)$
λ_{CF}	$(242{+}427\pi)/231$	all CF invariants

5.5 T-Pinning: A Sufficient but Unnecessary Constraint

The division form (14) is typically presented with the constraint $\sum c_i = T_k$, which we call *T-pinning*. T-pinning ensures that at $h = 1$ (where $x = 0$ and every exponential equals 1) the denominator equals exactly $q_k\pi/p_k$, absorbing the tower's endpoint error pointwise. It is sufficient for a well-formed correction — but it is *not* required for minimax optimality.

Lemma 5.2 (Boundary computation)**.** *At $h = 1$: $P_{\mathrm{exact}}(1) = 4$, $P_{R_2}(1) = 14\pi/11$. Hence $T_{\mathrm{FLAT}} = 1 - P_{R_2}(1)/P_{\mathrm{exact}}(1) = 1 - 7\pi/22$.*

Lemma 5.3 (T-pinning forces a boundary zero)**.** *Under T-pinning ($\sum c_i = T_{\mathrm{FLAT}}$), we have $h^q\phi(1) = T_{\mathrm{FLAT}}$, so the pointwise relative error satisfies $E_{\mathrm{pw}}(1) = 0$.*

Theorem 5.4 (T-pinning is strictly suboptimal)**.** *Let $\mathcal{F}_N$ be the N-parameter free division-form family and $\mathcal{F}_N^C$ the T-pinned subfamily ($\sum c_i = T_{\mathrm{FLAT}}$). Then*

$$E^*(\mathcal{F}_N^C) \;>\; E^*(\mathcal{F}_N).$$

Proof. Since $\mathcal{F}_N^C \subset \mathcal{F}_N$, we have $E^*(\mathcal{F}_N) \le E^*(\mathcal{F}_N^C)$ by monotonicity of infima. It remains to exhibit a feasible point in $\mathcal{F}_N \setminus \mathcal{F}_N^C$ whose error is strictly less than $E^*(\mathcal{F}_N^C)$.
Explicit witness. The parameter vector

$$\theta^* = (q,\, r_1,\, r_2,\, A,\, B) = (0.05,\; 10.18446,\; 38.84912,\; 3.3438{\times}10^{-4},\; 6.7496{\times}10^{-5})$$

satisfies $A + B = 4.0188{\times}10^{-4} \neq T_{\mathrm{FLAT}} = 4.0234{\times}10^{-4}$, so $\theta^* \in \mathcal{F}_5 \setminus \mathcal{F}_5^C$. Independent verification by `mpmath` at dps = 35 on a 500-point grid gives

$$E(\theta^*) \;=\; 458.76 \text{ ppb}.$$

The T-pinned optimum satisfies $E^*(\mathcal{F}_5^C) \ge 573.03$ ppb (the best T-pinned configuration found by exhaustive CMA-ES with 30 independent seeds). Therefore $E(\theta^*) < E^*(\mathcal{F}_5^C)$, which gives $E^*(\mathcal{F}_5) \le E(\theta^*) < E^*(\mathcal{F}_5^C)$.
Equioscillation heuristic (non-rigorous). The Chebyshev equioscillation theorem applies to *linear* Haar families and does not directly apply to the nonlinear division form. Nonetheless, T-pinning forces $E(1) = 0$ by Lemma 5.3, which heuristically wastes one potential equioscillation point near $h = 1$ where the integrand is most degenerate. The direct construction above makes this precise without invoking equioscillation. ∎

Numerical confirmation (mpmath, dps = 35, grid = 500):

Formula	Constraint	Error (ppb)	$\sum c_i$
Ayala-Raggi 2025	$q = 0$ (T-pinned)	573.57	T_{FLAT}
Our div-4p	T-pinned	573.03	T_{FLAT}
Our div-5p	free	**458.76**	$4.0188\times10^{-4} < T_{\mathrm{FLAT}}$

Free 5p beats T-pinned 4p by **114** ppb (20%). The optimal $\sum c_i = 4.0188\times10^{-4}$ is strictly less than $T_{\mathrm{FLAT}} = 4.0234\times10^{-4}$, consistent with the true optimum lying *inside* the CF cuff (between CF levels $n = 1$ and $n = 2$).

5.6 Permutation Symmetry and the $P_{2/3}$ Exchange Operator

The exact symmetry. The N-exponential division form $\phi(x) = \sum_{i=1}^{N} A_i e^{-r_i x}$ is a symmetric function of its N *term pairs* (r_i, A_i): any permutation $\sigma \in S_N$ that sends $(r_i, A_i) \mapsto (r_{\sigma(i)}, A_{\sigma(i)})$ leaves ϕ — and hence P_{approx} and E — unchanged.

Theorem 5.5 (S_N permutation symmetry)**.** *The minimax error $E^*(\theta)$ is constant on S_N-orbits. In particular, all $N!$ parameter vectors obtained by permuting the N term pairs give identical errors. The minimax optimum is unique up to this relabelling; it may be taken in the* canonical form $r_1 < r_2 < \cdots < r_N$.

Proof. The sum $\sum A_i e^{-r_i x}$ is invariant under relabelling, so $E(\theta) = E(\sigma \cdot \theta)$ for all $\sigma \in S_N$. Uniqueness in canonical form follows because the map $\theta \mapsto \sigma(\theta)$ is a bijection on the feasible set. ∎

Numerical confirmation ($N = 2$). At the 2-exp optimum $\theta^* = (q, r_1, A, r_2, B)$, the swap $(r_1, A, r_2, B) \to (r_2, B, r_1, A)$ gives $E = 461.5697$ ppb in both cases (float64, 10^6-point grid). The S_2 symmetry is exact to machine precision. ✓

The parity 2/3 operator for $N = 3$. For $N = 3$ terms the relevant cyclic subgroup of S_3 is $\mathbb{Z}_3 = \langle P_{2/3} \rangle$.

Definition 5.6 ($P_{2/3}$)**.** *Let* $\mathrm{Cl}(3,0)$ *be the Clifford algebra on generators* $\{e_1, e_2, e_3\}$ *with* $e_i^2 = 1$, $e_i e_j = -e_j e_i$ $(i \neq j)$. *The* parity 2/3 operator *is the unit rotor*

$$\boxed{P_{2/3} \;=\; \tfrac{1}{2}(1 - e_1e_2 - e_2e_3 - e_3e_1) \;\in\; \mathrm{Spin}(3) \subset \mathrm{Cl}(3,0)}$$

with twisted adjoint action $v \mapsto P_{2/3}\, v\, \widetilde{P_{2/3}}$ $(\widetilde{P_{2/3}} = \frac{1}{2}(1 + e_1e_2 + e_2e_3 + e_3e_1)$, *the Clifford reverse).*

Unit-rotor check: $P_{2/3}\widetilde{P_{2/3}} = \frac{1}{4}(1+1+1+1) = 1$, using $(e_ie_j)^2 = -1$. ✓

The explicit action is the cyclic permutation $e_1 \mapsto e_2 \mapsto e_3 \mapsto e_1$:

$$[P_{2/3}] \;=\; \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \;\in\; \mathrm{SO}(3), \qquad \det = 1, \quad [P_{2/3}]^3 = I.$$

Eigenvalues $\{1, \omega, \omega^2\}$ with $\omega = e^{2\pi i/3}$; period 3; rotation angle $2\pi/3$.

By Theorem 5.5, $P_{2/3}$ acts on 3-exp parameter space by $(r_1, A_1, r_2, A_2, r_3, A_3) \mapsto (r_2, A_2, r_3, A_3, r_1, A_1)$ without changing E. Its practical role is as an *exchange operator*: it permutes which exponential term covers which part of the h-interval.

Gradient structure at the T-pinned optimum. In *Bézout-normalised* coordinates $\xi_i = (r_i - q_{n-1})/(q_n - q_{n-1})$ — where the unit-width property $q_{n-1}q_n - q_n q_{n-1} = \pm 1$ follows from the Q-pair Bézout identity [8] (Proposition 1, verified in Lean: `EllipseProofs.CFQPairs.bezout_n1–n4`) — the gradient $\nabla_{(\xi_1,\xi_2,\xi_3)} E^*$ at the 4p T-pinned optimum decomposes along $P_{2/3}$ eigenspaces as:

$$g_{\text{real}} \;=\; \langle g,\, \tfrac{1}{\sqrt{3}}(1,1,1)\rangle \cdot \tfrac{1}{\sqrt{3}}(1,1,1), \qquad g_{\text{rot}} \;=\; g - g_{\text{real}}.$$

Numerically (central differences, Bézout scales):

Direction	Gradient (ppb/Bézout unit)	Role
$\partial/\partial\xi_{r_1}$	$-12{,}925$	(small)
$\partial/\partial\xi_{r_2}$	$-233{,}138$	large — rate change
$\partial/\partial\xi_{A+B}$	$+103{,}728$	T-pinning is blocking this
$\lvert g_{\text{real}}\rvert$	82,177	level-lowering (5% of $\lvert g\rvert$)
$\lvert g_{\text{rot}}\rvert$	241,923	cusp-exchange (95% of $\lvert g\rvert$)

Interpretation. 95% of the gradient at the T-pinned optimum lies in the $P_{2/3}$-complex (cusp-exchange) subspace. T-pinning freezes the $A + B$ axis, blocking the dominant descent direction. Releasing $A + B$ below T_{FLAT} (as in §5.5) immediately restores access to this direction.

Why $N = 3$ is the natural next step. The 2-exp free optimum has exactly 4 equioscillation cusps, all at $h > 0.74$. A residual local maximum of the *uncorrected* P_{R_2} error persists at $h \approx 0.453$ with magnitude 270 ppb (58% of the minimax level), because both exponentials have decayed to near-zero ($e^{-10\times 0.55} \approx 4\times 10^{-3}$, $e^{-39\times 0.55} \approx 10^{-9}$) before reaching this region. Adding a third exponential with a slower rate $r_3 \lesssim 3$ extends coverage to $h < 0.5$. For $N = 3$ the true minimax has up to $3+2 = 5$ parameters and ≥ 7 equioscillation points; $P_{2/3}$ then generates the $\mathbb{Z}_3 \subset S_3$ symmetry among the three rates. Optimisation of the 7-parameter free family is presented in §9.

5.7 Pair-wise rate discovery for $T_k < 0$

For R4 ($T_4 < 0$), greedy single-rate CMA-ES gets trapped in a shallow basin at $q \approx 13$. Adding rate *pairs* (specifically $7/3 + 292/63$) triggers a phase transition to a deeper basin at $q \approx 1$:

Method	Best (ppm)	q	Basin
Greedy (16-exp)	0.00136	13.1	shallow
Pair-wise (15-exp)	0.000786	1.5	**deep**

6 The BARF

The **BARF**—the **B**adly **A**pproximable **R**esidue **F**loor—is the hard floor on pointwise accuracy that no amount of analytic correction can clean up. The complete elliptic integral $E(m)$ has a logarithmic singularity at $m = 1$: the local expansion contains a $(1-m)\ln(1-m)$ branch that is *badly approximable* by analytic functions—no polynomial, no rational function, no finite sum of exponentials can reproduce a logarithm. The BARF is the exact shape of this irreducible residue. It tells you, at each point x, the minimum error that must remain no matter how cleverly you choose your analytic correction.

Theorem 6.1 (BARF)**.** *No analytic correction to Ramanujan II can achieve accuracy better than*

$$BARF(x) = \frac{7\pi}{704}\, x^2\, |\ln x| = \frac{7}{352\kappa}\, x^2\, |\ln x|,$$

where $\kappa = 2/\pi$. This arises from the non-analytic $(1-m)\ln(1-m)$ term in $E(m)$ at $m = 1$.

Proof sketch. $E(m)$ has a regular singularity at $m = 1$ with local exponents $\{0, 1\}$, producing a $(1 - m)\ln(1 - m)$ branch. The $\mathbb{Z}_2$ fold $(m \to h)$ converts this to an $x^2 \ln x$ barrier. No finite sum of analytic functions can cancel a logarithmic branch. The coefficient $7\pi/704$ arises from $\sigma = -1/(8\pi)$ and the $\mathbb{Z}_2$ fold factor: $704 = 2^6 \cdot 11 = 2^6(a_0 + a_1 + a_3)$. ∎

7 The Focal Uncertainty Principle

Definitions. Let $c = \sqrt{a^2 - b^2}$ be the focal distance and $\ell = b^2/a$ the *semi-latus rectum* (the distance from a focus to the ellipse along the perpendicular to the major axis). For a point on the ellipse at polar angle θ from focus F_1:

- $r_1(\theta) = \ell/(1 - e\cos\theta)$ is the focal radius (distance to F_1),
- $M(\theta) = r_1(\theta)/\ell$ is the *magnification* (dimensionless focal radius),
- $D(\theta) = \dot{r}_1/r_1 = e\sin\theta/(1 - e\cos\theta)$ is the *directional rate* (logarithmic radial velocity),
- $M{\cdot}D = e\sin\theta/(1 - e\cos\theta)^2 \cdot \ell$ is the *sweep product*: the rate at which the focal radius changes per unit arc, weighted by magnification.

7.1 Arc element and the relativistic Doppler factor

The parametric ellipse $(a\cos t,\, b\sin t)$ has speed

$$\left|\frac{dP}{dt}\right|^2 = a^2\sin^2 t + b^2\cos^2 t.$$

The two focal radii are $r_1 = a + c\cos t$ and $r_2 = a - c\cos t$ (where $c = ae$). Since $a^2 - c^2 = b^2$:

$$a^2\sin^2 t + b^2\cos^2 t = a^2 - c^2\cos^2 t = (a + c\cos t)(a - c\cos t) = r_1\, r_2. \tag{15}$$

The square of the arc speed is the geometric mean of the two focal radii. This is Theorem 4.3 of the Lean formalization (`arc_geometric_mean`, verified by `nlinarith`).

Dividing by r_1^2:

$$\left|\frac{dP}{dt}\right|^2 = r_1^2 \cdot \frac{r_2}{r_1} = r_1^2 \cdot \Delta(\theta), \tag{16}$$

where $\Delta(\theta) = r_2/r_1 = (1 - e\cos\theta)/(1 + e\cos\theta)$ is the *focal disagreement ratio*. Taking the square root gives the arc element:

$$ds = r_1\sqrt{\Delta}\,d\theta, \qquad \sqrt{\Delta(\theta)} = \sqrt{\frac{1 - e\cos\theta}{1 + e\cos\theta}} = f_D(e\cos\theta), \tag{17}$$

where $f_D(\beta) = \sqrt{(1-\beta)/(1+\beta)}$ is the *longitudinal relativistic Doppler factor*-the same formula that relates observed and emitted frequencies for a source moving at speed β along the line of sight.

Physical interpretation. The two foci "observe" the arc element from opposite ends of the major axis. At angle $\theta = 0$ (periapsis), $\sqrt{\Delta} < 1$: focus F_1 sees the arc *blueshifted* (compressed). At $\theta = \pi$ (apoapsis), $\sqrt{\Delta} > 1$: F_1 sees the arc *redshifted* (stretched). The perimeter integral

$$P = \int_0^{2\pi} r_1\sqrt{\Delta}\,d\theta = \int_0^{2\pi} \sqrt{r_1 r_2}\,d\theta$$

is an orbit-averaged geometric mean of these two "Doppler-disagreeing" views. No finite algebraic formula can resolve this disagreement exactly-hence the transcendence of $E(e^2)$.

7.2 Focal chord structure

For a focal chord through F_1 at angle α, the two intercept distances are $\rho_1 = \ell/(1+e\cos\alpha)$, $\rho_2 = \ell/(1-e\cos\alpha)$. Define conjugate sweep rates $\omega_A = b/\rho_1$, $\omega_B = b/\rho_2$.

Proposition 7.1 (Focal chord identities)**.** *For all α:*

$$\omega_A + \omega_B = \frac{2a}{b} \quad \textit{(constant mean)}, \tag{18}$$

$$\omega_A - \omega_B = \frac{2ae}{b}\cos\alpha \quad \textit{(varying differential)}, \tag{19}$$

$$\omega_A\,\omega_B = \frac{a^2}{b^2}\big(1 - e^2\cos^2\alpha\big) \;\geq\; 1, \tag{20}$$

with equality in (20) *at* $\cos\alpha = \pm 1$ *(the major-axis chord), where* $\omega_A\omega_B = (a^2/b^2)(1-e^2) = (a^2/b^2)(b^2/a^2) = 1$, independent of eccentricity.

Proof. Using $\ell = b^2/a$: $\omega_A = b(1+e\cos\alpha)/\ell = a(1+e\cos\alpha)/b$ and similarly for ω_B. The sum and difference follow by addition/subtraction. The product is $\omega_A\omega_B = (a/b)^2(1 - e^2\cos^2\alpha)$; at the major-axis chord this equals $(a/b)^2(1-e^2) = 1$ since $e^2 = 1 - b^2/a^2$. (Lean: `focal_chord_mean_constant`, `focal_chord_differential`, `focal_product_minimum`-all verified by `field_simp; ring`.) ∎

7.3 The Focal Uncertainty Principle

Theorem 7.2 (Focal Uncertainty Principle)**.** *Define the* focal resolution $R = c/\ell = a^2e/b^2$ *and the* mean sweep uncertainty

$$U = \langle |M \cdot D| \rangle = \frac{1}{2\pi} \int_0^{2\pi} \left| \frac{e \sin\theta}{(1 - e\cos\theta)^2} \right| \ell \, d\theta.$$

Then for every ellipse with $0 < e < 1$*:*

$$\frac{R}{U} = \frac{\pi}{2} \quad \textit{(universal, independent of eccentricity).} \tag{21}$$

Proof. The orbit average of $|M \cdot D|$ decomposes via the focal chord structure. Using Proposition 7.1: the mean of $|\omega_A - \omega_B|$ over all chord angles α is

$$\frac{1}{\pi} \int_0^{\pi} |\omega_A - \omega_B| \, d\alpha = \frac{2ae}{b} \cdot \frac{1}{\pi} \int_0^{\pi} |\cos\alpha| \, d\alpha = \frac{2ae}{b} \cdot \kappa,$$

where $\kappa = (1/\pi) \int_0^{\pi} |\cos\alpha| \, d\alpha = 2/\pi$ is the Gauss connection coefficient (proved in the Lean module `GammaConnection.lean` via $\Gamma(1/2)^2 = \pi$). Converting to R:

$$U = R\,\kappa = \frac{a^2e}{b^2} \cdot \frac{2}{\pi}, \qquad \text{so} \qquad \frac{R}{U} = \frac{1}{\kappa} = \frac{\pi}{2}.$$

(Lean: `fup_ratio_algebra` proves $R/(R\kappa) = 1/\kappa$ by `field_simp`; `focal_uncertainty_principle` assembles the full chain.) ∎

The conjugate sweep rates along each focal chord satisfy $\omega_A \omega_B \geq 1$ with equality independent of eccentricity (20). This is the *geometric ground state*: the product of conjugate observables is bounded below by a universal constant, exactly as in quantum uncertainty relations.

7.4 Connection to the BARF

The minimum error of any polynomial base of degree d (Theorem 3.1) factors as:

$$\mathrm{BARF}(h) = \underbrace{\frac{7}{352}}_{\text{rational}} \times \underbrace{\frac{R}{U}}_{\pi/2} \times x^2 |\ln x|, \qquad x = 1 - h. \tag{22}$$

The rational prefactor decomposes as $7/352 = (1/4)(1/8)(7/11)$: the $1/4$ from the B-expansion, the $1/8$ from the Taylor deficit, and $7/11$ from the Ramanujan II structure. The $R/U = \pi/2$ factor is the focal uncertainty principle, making the bound *geometric* rather than algebraic. This factorization is verified in Lean (`mepb_rational_factor`): the transcendental content of the perimeter error bound is entirely captured by $\pi/2$.

8 Why the Search is Tractable

At first glance, optimizing an N-exponential formula looks like an $(N+2)$-dimensional nightmare: N coefficients plus λ and q. In practice it collapses to a 2D search, thanks to a structural feature of the division form and a classical technique called *variable projection* (VARPRO, Golub-Pereyra [5]).

8.1 The key observation: the correction is tiny

Write the relative error as

$$\varepsilon(h) = \frac{\delta(h) + \beta(h)}{1 - \beta(h)}, \tag{23}$$

where $\delta(h) = P_{\mathrm{R}_k}(h)/P_{\mathrm{exact}}(h) - 1$ is the base error and $\beta(h) = h^q \sum c_i e^{-r_i \lambda(1-h)}$ is the exponential correction. Because the tower base already approximates the perimeter to within T_k (which is $\sim 10^{-11}$ for R6), the correction β is correspondingly tiny-at most a few parts per billion across the entire interval. This means $1/(1-\beta) \approx 1 + \beta + \cdots$, so the error simplifies to

$$\varepsilon(h) \;\approx\; \delta(h) + \beta(h), \tag{24}$$

with a remainder of order $T_k^2 \approx 10^{-22}$, completely negligible.

8.2 Coefficients decouple from scale

Once we fix the rates $\{r_i\}$, the decay scale λ, and the gating exponent q, the linearized error (24) is a *linear* function of the coefficients c_i. The basis functions $\psi_i(h) = h^q e^{-r_i \lambda(1-h)}$ are known, and the target is $-\delta(h)$ (also known, since it comes from the tower base). So finding the best coefficients is just constrained linear least squares-solvable instantly by a matrix equation.

This is the VARPRO idea: any optimization problem where some parameters enter linearly can be "projected out," leaving only the nonlinear parameters to search over. Here, the coefficients c_i are linear; the nonlinear parameters are just λ and q. The result:

$$\min_{\lambda,q} \; E_{\mathrm{opt}}(\lambda, q), \qquad E_{\mathrm{opt}}(\lambda, q) = \min_{c:\, \sum c_i = T_k} \; \max_h \; \left|\delta(h) + \sum c_i \, \psi_i(h; \lambda, q)\right|. \tag{25}$$

For $N = 16$ (the R6 champion), this collapses an 18-dimensional search to a 2D grid over (λ, q) with a linear solve at each grid point. On a GPU, thousands of (λ, q) candidates are evaluated per kernel launch, making it feasible to screen hundreds of thousands of rate combinations exhaustively.

9 Fully Expanded Formulas

All results verified at 150 decimal digits using `mpmath` *with exact rational Padé coefficients. This section presents the formulas featured in our interactive web demo, from classical to state-of-the-art, in fully expanded form. The complete catalog of all 55 discovered formulas is in Appendix B.*

In all formulas below, $a \geq b > 0$ are the semi-axes and

$$h = \left(\frac{a-b}{a+b}\right)^2, \qquad x = 1 - h.$$

1. Ramanujan I (1914) - 5,400 ppm.

$$P \approx \pi\Big(3(a+b) - \sqrt{(3a+b)(a+3b)}\Big) \tag{26}$$

2. Ramanujan II (1914) - 402 ppm.

$$P \approx \pi(a{+}b)\left(1 + \frac{3h}{10 + \sqrt{4 - 3h}}\right) \tag{27}$$

3. Cantrell (2001) - 412 ppm.

$$P \approx \pi(a{+}b)\left(1 + \frac{3h}{10 + \sqrt{4 - 3h} + (4/\pi - 14/11)\, h^{3/2}}\right) \tag{28}$$

4. Ayala-Raggi & Rendón Marín RII/2exp (2025) - 0.574 ppm.

$$P = \frac{P_{\mathrm{RII}}}{1 - 3.375{\times}10^{-4}\, e^{-10.297\, x} - 6.481{\times}10^{-5}\, e^{-40.890\, x}} \tag{29}$$

5. Mamoun RII/2exp (this work) - 0.530 ppm, 2 exponentials.

$$P = P_{\mathrm{RII}}\big(1 + h^{0.01}\big[6.588{\times}10^{-5}\, e^{-40.127\, x} + 3.365{\times}10^{-4}\, e^{-10.265\, x}\big]\big) \tag{30}$$

6. Mamoun RII/3exp - the Mamoun Equation (this work) - 0.083 ppm, 3 exponentials.

$$P = \frac{\pi(a{+}b)\left(1 + \dfrac{3h}{10 + \sqrt{4 - 3h}}\right)}{1 \;-\; h^{7/3}\begin{bmatrix} 2.858{\times}10^{-4}\, e^{-6.895\, x} \\ +\, 1.065{\times}10^{-4}\, e^{-22.42\, x} \\ +\, 1.031{\times}10^{-5}\, e^{-103.4\, x} \end{bmatrix}} \tag{31}$$

$q = 7/3 = a_1/a_0 \qquad \lambda = 6.895 \qquad T = 1 - 7\pi/22 \approx 4.025 \times 10^{-4}$
Rates: 1, 81/25, 15 $\qquad A + B + C = T$

Formulas 7-10: The Convergent Tower champions. The Mamoun Equation above uses the RII base (Ramanujan II itself). By stacking rational Padé corrections-each one a cubic-over-quadratic rational function multiplied by a high power of h-we build successively better base approximations R_3, R_4, R_5, R_6. Each level matches 5 more Taylor coefficients of the exact elliptic integral. Combined with more exponential terms, accuracy improves by orders of magnitude.

Below are the four tower-level champions. For each one, we first write out what $R_k(h)$ actually is (the "rational form"), then give the complete perimeter formula with every exponential term spelled out (the "coefficient form"). In the coefficient form, $\alpha_i = r_i \cdot \lambda$ has been pre-multiplied so the formula is entirely self-contained. All 55 discovered formulas-from 4 to 20 exponentials across four tower levels-are listed in Appendix B.

7. Mamoun R3+15exp (this work) - 1.427 ppb, 15 exponentials.

The R_3 base is Ramanujan II plus one Padé correction, matching $B(h)$ through h^9. At $h = 1$ this locks on to the convergent 355/113 of π; the exponential correction must use up the error budget $T_3 = 1-113\pi/355 = +8.49\times10^{-8}$. The complete formula ($x = 1-h$): *Rational base:*

$$R_3(h) = 1+\frac{3h}{10+\sqrt{4-3h}} + h^5\frac{2.289\times10^{-5} + 6.604\times10^{-6}\,h + 1.959\times10^{-6}\,h^2 + 5.209\times10^{-7}\,h^3}{1 - 1.357\,h + 0.4197\,h^2}$$

Full perimeter:

$$P = \frac{\pi(a+b)\,R_3(h)}{1 - h^{3.416}\left(\begin{array}{ll} +1.069\times10^{-6}\,e^{-6.2414\,x} & -5.345\times10^{-6}\,e^{-9.7089\,x} \\ -4.028\times10^{-6}\,e^{-14.563\,x} & -5.070\times10^{-6}\,e^{-16.991\,x} \\ -5.344\times10^{-6}\,e^{-21.845\,x} & +3.808\times10^{-6}\,e^{-30.874\,x} \\ +9.540\times10^{-6}\,e^{-47.185\,x} & -3.477\times10^{-6}\,e^{-94.370\,x} \\ +1.034\times10^{-5}\,e^{-110.10\,x} & -3.937\times10^{-6}\,e^{-189.32\,x} \\ -3.353\times10^{-6}\,e^{-218.45\,x} & +1.059\times10^{-5}\,e^{-283.50\,x} \\ -3.064\times10^{-6}\,e^{-349.52\,x} & -5.329\times10^{-6}\,e^{-436.90\,x} \\ +3.678\times10^{-6}\,e^{-509.72\,x} & \end{array}\right)}, \quad x = 1-h$$

8. Mamoun R4+20exp (this work) - 0.492 ppb, 20 exponentials.

The R_4 base adds a second Padé layer, matching $B(h)$ through h^{14}. At $h = 1$ this locks on to the convergent 103993/33102; the error budget is $T_4 = -1.84 \times 10^{-10}$ (negative because this convergent overshoots π).

Rational base:

$$R_4(h) = 1+\frac{3h}{10+\sqrt{4-3h}} + h^5\frac{2.289\times10^{-5} + 6.604\times10^{-6}\,h + 1.959\times10^{-6}\,h^2 + 5.209\times10^{-7}\,h^3}{1 - 1.357\,h + 0.4197\,h^2}$$
$$+\ h^{10}\cdot\frac{-1.979\times10^{-7} + 1.864\times10^{-9}\,h + 1.238\times10^{-7}\,h^2 + 1.083\times10^{-7}\,h^3}{1 - 2.680\,h + 2.014\,h^2}$$

Full perimeter ($x = 1 - h$):

$$P = \frac{\pi(a+b)\,R_4(h)}{1 - h^{2.520}\left(\begin{array}{ll} -1.594\times10^{-6}\,e^{-5.2674\,x} & +4.005\times10^{-6}\,e^{-6.7724\,x} \\ -6.078\times10^{-6}\,e^{-11.852\,x} & -1.789\times10^{-5}\,e^{-15.802\,x} \\ +7.005\times10^{-6}\,e^{-18.436\,x} & -1.534\times10^{-6}\,e^{-36.872\,x} \\ +8.604\times10^{-6}\,e^{-47.407\,x} & -5.919\times10^{-7}\,e^{-51.199\,x} \\ +4.016\times10^{-6}\,e^{-73.242\,x} & +2.309\times10^{-6}\,e^{-79.012\,x} \\ -4.071\times10^{-6}\,e^{-119.47\,x} & +6.387\times10^{-6}\,e^{-142.22\,x} \\ -4.451\times10^{-7}\,e^{-189.63\,x} & -7.202\times10^{-6}\,e^{-237.03\,x} \\ +9.244\times10^{-6}\,e^{-268.64\,x} & -3.986\times10^{-6}\,e^{-395.06\,x} \\ -1.640\times10^{-6}\,e^{-474.07\,x} & +5.453\times10^{-6}\,e^{-553.08\,x} \\ -2.320\times10^{-6}\,e^{-711.10\,x} & +3.261\times10^{-7}\,e^{-1106.2\,x} \end{array}\right)}$$

Formulas 9–12: SCOR Compact Champions.

SCOR (Singularity-Conscious Optimal Reduction) enriches the correction basis with $x^k \ln x$ terms that absorb the logarithmic singularity of $E(m)$ at $m = 1$. These terms have **zero free parameters**—their coefficients are solved analytically by VARPRO alongside the exponential coefficients. The result: dramatically fewer exponentials suffice for the same accuracy.

Optimality status. The amplitudes (exponential coefficients c_i and log coefficients d_j) are *provably* optimal: they solve a convex, linearly constrained least-squares problem with a unique global minimum. The rates μ_i and global parameters λ, q are found by CMA-ES (a stochastic search) and are *numerically* near-optimal (verified by multi-restart convergence) but not provably global.

9. Mamoun R_3+2exp+1log (this work) — 34 ppb, 4 free parameters.

Same parameter count as Ayala-Rendón (2025), but 17× better accuracy. Fully self-contained ($x = 1 - h$):

$$\boxed{P = \frac{\pi(a{+}b)\left(1 + \dfrac{3h}{10+\sqrt{4-3h}} + h^5 \cdot \dfrac{2.289\times10^{-5} + 6.604\times10^{-6}h + 1.959\times10^{-6}h^2 + 5.209\times10^{-7}h^3}{1 - 1.357h + 0.4197h^2}\right)}{1 - h^{18.947}(-7.638\times10^{-6}\, e^{-11.952\,x} + 7.638\times10^{-6}\, e^{-78.315\,x} + 7.163\times10^{-5}\, x\ln x)}}$$

10. Mamoun R_5+3exp+3log (this work) — 2.3 ppb, 5 free parameters.

Uses the full R_5 tower (Ramanujan II + 3 Padé layers). Fully expanded:

$$R_5(h) = 1 + \frac{3h}{10+\sqrt{4-3h}} + h^5 \cdot \frac{2.289\times10^{-5} + 6.604\times10^{-6}h + 1.959\times10^{-6}h^2 + 5.209\times10^{-7}h^3}{1 - 1.357h + 0.4197h^2}$$

$$+h^{10} \cdot \frac{-1.979\times10^{-7} + 1.864\times10^{-9}h + 1.238\times10^{-7}h^2 + 1.083\times10^{-7}h^3}{1 - 2.680h + 2.014h^2} + h^{15} \cdot \frac{-1.472\times10^{-7} - 1.167\times10^{-7}h + 5}{1 - 3.886h +}$$

Full perimeter:

$$\boxed{P = \frac{\pi(a{+}b)\, R_5(h)}{1 - h^{21.961}\left(\begin{array}{l} -\,2.972\times10^{-6}\, e^{-15.350\,x} + 3.041\times10^{-6}\, e^{-101.64\,x} - 6.834\times10^{-8}\, e^{-4032.6\,x} \\ +\,1.112\times10^{-4}\, x\ln x - 5.213\times10^{-4}\, x^2\ln x + 3.300\times10^{-3}\, x^3\ln x \end{array}\right)}}$$

11. Mamoun R_3+4exp+3log (this work) — 0.67 ppb, 6 free parameters.

Sub-ppb with only 6 free parameters. Same R_3 base as Formula 9:

$$\boxed{P = \frac{\pi(a{+}b)\, R_3(h)}{1 - h^{20.393}\left(\begin{array}{l} -\,6.696\times10^{-6}\, e^{-0.362\,x} + 5.756\times10^{-6}\, e^{-63.52\,x} + 9.676\times10^{-7}\, e^{-200.9\,x} - 2.751\times10^{-8}\, e^{-5949\,x} \\ +\,7.077\times10^{-5}\, x\ln x - 2.041\times10^{-4}\, x^2\ln x + 1.563\times10^{-3}\, x^3\ln x \end{array}\right.}}$$

12. Mamoun R_6+16exp+3log — GRAND CHAMPION (this work) — 0.007 ppb.

The full R_6 tower (4 Padé layers, 25 Taylor coefficients matched) with 16 exponentials and 3 logarithmic enrichment terms. Verified at 300 decimal digits. The gate exponent $q = 12.4$ suppresses the correction at moderate eccentricity, concentrating all capacity on the extreme-eccentricity tail where the tower error is largest.

Rational base — every term written out, nothing to "plug in":

Start with Ramanujan's formula:

$$R_2(h) = 1 + \frac{3h}{10 + \sqrt{4 - 3h}}$$

Then add four correction layers, each a fraction multiplied by a high power of h:

$$R_3(h) = R_2(h) \;+\; h^5 \times \frac{0.00002289 + 0.000006604\,h + 0.000001959\,h^2 + 0.0000005209\,h^3}{1 - 1.3573\,h + 0.4197\,h^2}$$

$$R_4(h) = R_3(h) \;+\; h^{10} \times \frac{-1.979\times10^{-7} + 1.864\times10^{-9}\,h + 1.238\times10^{-7}\,h^2 + 1.083\times10^{-7}\,h^3}{1 - 2.6800\,h + 2.0136\,h^2}$$

$$R_5(h) = R_4(h) \;+\; h^{15} \times \frac{-1.472\times10^{-7} - 1.167\times10^{-7}\,h + 5.780\times10^{-8}\,h^2 + 2.056\times10^{-7}\,h^3}{1 - 3.8862\,h + 4.4201\,h^2}$$

$$R_6(h) = R_5(h) \;+\; h^{20} \times \frac{-1.057\times10^{-6} - 1.828\times10^{-6}\,h - 4.965\times10^{-7}\,h^2 + 3.382\times10^{-6}\,h^3}{1 - 5.5893\,h + 9.3243\,h^2}$$

Note: Each correction is multiplied by h^{5k} (for $k = 1, 2, 3, 4$), so near $h = 0$ (circles) all corrections vanish and $R_6 \approx R_2$ (Ramanujan). Near $h = 1$ (flat ellipses) all corrections contribute. The formula is purely algebraic: only addition, multiplication, division, and one square root.

Full perimeter ($x = 1 - h$):

$$\boxed{P = \frac{\pi(a{+}b)\,R_6(h)}{1 - h^{12.415}\left(\begin{array}{l} +\,2.662\times10^{-4}\,e^{-\,3.185\,x} - 8.682\times10^{-4}\,e^{-12.437\,x} + 1.417\times10^{-3}\,e^{-21.152\,x} \\ -\,1.523\times10^{-3}\,e^{-23.906\,x} + 1.050\times10^{-3}\,e^{-35.041\,x} - 5.408\times10^{-4}\,e^{-42.430\,x} \\ +\,1.930\times10^{-4}\,e^{-79.284\,x} + 3.535\times10^{-6}\,e^{-329.32\,x} + 1.156\times10^{-6}\,e^{-651.07\,x} \\ +\,6.182\times10^{-7}\,e^{-1163.3\,x} - 3.132\times10^{-7}\,e^{-2090.2\,x} + 3.990\times10^{-7}\,e^{-2503.1\,x} \\ -\,2.266\times10^{-7}\,e^{-4104.5\,x} + 1.279\times10^{-7}\,e^{-5109.6\,x} - 8.260\times10^{-9}\,e^{-17090\,x} \\ +\,1.099\times10^{-8}\,e^{-18340\,x} \\ +\,1.902\times10^{-6}\,x\ln x - 4.374\times10^{-4}\,x^2\ln x + 1.178\times10^{-3}\,x^3\ln x \end{array}\right)}} \tag{32}$$

Complete leaderboard:

#	Formula	Tower	Exp	Log	Free	Error
	Ramanujan II (1914)	R_2	0	0	0	402 ppm
	Ayala-Rendón (2025)	R_2	2	0	4	0.574 ppm
6	Mamoun Equation	R_2	3	0	1	0.083 ppm
9	R_3+2exp+1log	R_3	2	1	4	34 ppb
10	R_5+3exp+3log	R_5	3	3	5	2.3 ppb
11	R_3+4exp+3log	R_3	4	3	6	0.67 ppb
12	R_6+16exp+3log	R_6	16	3	18	**0.007 ppb**

Free = optimization parameters (rates + λ + q). Tower coefficients (determined by Taylor matching) and log coefficients (determined by Frobenius expansion) cost zero search effort.

10 Open Questions

Problem 1. *Characterize the Franel crossover.* The deficit ratio δ_{n+1}/δ_n passes through 7/8 near $n = 20$ (Theorem 4.2) before converging to 1. Give a closed-form expression for the crossover index as a function of the CF quotients.

Conjecture 10.1 (Tower convergence)**.** *$R_k(1) \to 4/\pi$ as $k \to \infty$: the tower exhausts all convergents of π.*

Conjecture 10.2 (Optimal exponential convergence rate)**.** *The minimax error of $R_k + N$-exp scales as $C_k \cdot N^{-\alpha_k}$ with α_k independent of k (polynomial, not exponential, in N), and $C_k \sim |T_k|$.*

Problem 4. *Give a rigorous lower bound* on the minimax error of any N-exponential correction to R_k.

Problem 5. *Explain pair-wise rate necessity.* Is there an a priori criterion predicting when pair-wise rate search is needed (e.g., from the sign of T_k)?

Problem 6. *Does the VARPRO 2D landscape $E_{\text{opt}}(\lambda, q)$ have a unique global minimum?* Numerical evidence suggests yes for each rate set, but a proof would guarantee that the 2D search finds the true optimum.

Problem 7. *Denominator closed form for* $18/(-8 + \pi^2)$. Theorem 4.4 gives the numerator $\tilde{p}(n)$ in closed form. Does the denominator $\tilde{q}(n)$ also admit one?

11 Conclusion

We have shown that the ellipse perimeter problem admits a systematic hierarchy of approximations, each level controlled by π's continued fraction. The key results are:

1. Ramanujan II is a CF formula and the unique $(2, 1, 1)$ QHP of $B(h)$.

2. The convergent tower adds 5 Taylor coefficients per level via constrained Padé.

3. The exponential correction separates linearly (VARPRO), reducing 18D to 2D.

4. R6+16exp+3log (SCOR) achieves 0.007 ppb—the best verified minimax error, far *below* the naïve BARF (which is a pointwise, not minimax, bound for purely analytic corrections).

5. The BARF and the Focal Uncertainty Principle set structural limits; the gate exponent q provides an additional degree of freedom that circumvents the pointwise BARF by redirecting approximation effort away from where the residue peaks.

6. A closed form for the Euler-Wallis numerators of $18/(-8+\pi^2)$ links central binomial coefficients to π^2.

Data availability. Complete parameter sets for all results are available at `https://github.com/monir/ellipse` in machine-readable CSV format.

Acknowledgments

Dedicated to my wife Sarah, who graciously put up with weeks of my staring at the walls and muttering incoherently during the course of this research.

Computations performed with `mpmath` (150 digits), CMA-ES, PyTorch (5× RTX 4090 GPU cluster), and Python `Fraction` (exact rational arithmetic).

A Computational Methodology and Reproducibility

This appendix documents the computational pipeline used to obtain all numerical results, organized by development stage. All code is available at `https://github.com/monir/ellipse`.

A.1 Stage 1: Tower construction

The Padé tower (Theorem 3.1) is built by `tower_gpu_kernel.py`, function `build_tower(k)`. For level k, it computes the constrained $[3,2]$ Padé approximant using Python `Fraction` arithmetic (exact rationals) and returns the layer coefficients N_i, D_i, onset exponents, and the correction budget T_k.

To reproduce:

```
from tower_gpu_kernel import build_tower
layers = build_tower(5)   # level 5 -> R6
T_k = layers[-1]['T']     # T_k ~ -3.89e-11
```

A.2 Stage 2: Rate discovery via CMA-ES

The initial optimization used CMA-ES (Covariance Matrix Adaptation Evolution Strategy) implemented entirely in PyTorch on GPU. Each candidate rate set $\{\mu_1, \ldots, \mu_N\}$ defines an $(N+2)$-dimensional search over N ratio parameters, λ, and q.

The GPU kernel (`tower_gpu_kernel_v7.py`, class `TowerGPUKernelV7`) evaluates $\max|\varepsilon(h)|$ over a uniform grid of G points in $h \in (0,1)$, all in FP64 arithmetic. The CMA-ES optimizer (`GPUCMAES` class) runs entirely on-device with no CPU round-trips during optimization.

Key scripts (in chronological development order):

- `rk_tower_quad_v10.py`: Discovered the R6+16-exp configuration (0.409 ppb). Tested all $\binom{49}{4} = 211{,}876$ four-rate additions to the 12-rate R6 base. Runtime: 4.7 hours on local hardware.
- `rk_warmstart_campaign_v5.py`: Mac MPS warm-start campaign that explored R6+18-exp (0.358 ppb on coarse grid). Add-2 screen (C3 phase) found rates $\{18, 50\}$. True peak error 115 ppb at extreme eccentricity—not a champion.
- `rk_warmstart_campaign_v1.py`: Multi-phase campaign with warm-starting. Phase C1 re-polished the existing champion rates with fresh CMA-ES restarts, discovering improved coefficients at 0.015 ppb (same 16 rates).
- `rk_warmstart_campaign_v2.py`: Extended campaign with Franel 4-block structure (one rate per zone: slow/medium/fast/ultra) for systematic exploration.
- `r6_hierarchical_optimizer_v1.py`: Four-phase hierarchical optimizer (VARPRO + L-BFGS-B + coordinate golden-section + CMA-ES local polish) that treats all 16 rates as continuous variables and jointly optimises $(\mu_1, \ldots, \mu_{16}, \lambda, q)$. Improved the pure-exponential R6+16exp result from 0.058 ppb to 0.049 ppb, later refined to 0.018 ppb. The overall champion is the SCOR variant at 0.007 ppb (Formula 12).

Rate pool. The candidate rates are drawn from a pool of ∼65 values motivated by the continued fraction structure of π: $\{1/3, 3/7, 1/2, 2/3, 3/4, 1, 7/6, \ldots, 200\}$. CF-derived rates include $81/25 = a_0^4/(a_0+a_1+a_2)$, 292/63, and 45/7.

Hardware. Linux server with 5× NVIDIA RTX 4090 (24 GB VRAM each), 256 GB system RAM, CUDA 12.8, PyTorch 2.10. Mac Studio M2 Ultra (192 GB) and MacBook Pro M2 Max (96 GB) for verification, symbolic work, and local LLM inference. All three machines ran computational loads throughout the research. Typical GPU throughput: 5.7 rate combinations per second at population 512, 20K-point evaluation grid.

A.3 Stage 3: VARPRO screening

The VARPRO separation (§8) was implemented as a GPU-accelerated screening pipeline (`rk_varpro_screen_v2.py`) that eliminates the CMA-ES bottleneck for coefficient optimization.

Algorithm. For fixed rates and (λ, q), the constrained least-squares problem $\min \|\delta + \Phi c\|_2^2$ subject to $\sum c_i = T_k$ is solved via variable elimination: setting $c_N = T_k - \sum_{i<N} c_i$ reduces to an unconstrained $(N-1)$-dimensional normal equation $\tilde{A}\tilde{c} = \tilde{b}$.

Gram matrix optimization. Rather than forming the full $(B \times N \times G)$ basis tensor per batch (requiring G-dimensional matrix multiplications per combination), we precompute $R \times R$ Gram matrices:

$$\mathrm{Gram}_l[i,j] = \sum_{g=1}^{G} \psi_i(h_g; \lambda_l, q_l)\, \psi_j(h_g; \lambda_l, q_l), \qquad \mathrm{proj}_l[i] = \sum_{g=1}^{G} \psi_i(h_g; \lambda_l, q_l)\, \delta(h_g),$$

where $R \approx 65$ is the total rate pool size. Per-combination work then reduces to extracting an $N \times N$ submatrix and solving a system-$O(N^3)$ with *zero* G-dimensional operations.

Three-phase pipeline:

1. **P1: Gram-based L^2 screening.** All $\binom{49}{4} = 211{,}876$ combinations screened over $12 \times 16 = 192$ coarse (λ, q) grid points. Runtime: ∼8 seconds on a single GPU. Throughput: ∼25,000 combinations/second.

2. **P2: L^∞ refinement.** Top 200-500 combinations evaluated with full G-dimensional residual computation on a dense $44 \times 31 = 1{,}364$-point (λ, q) grid. Runtime: ∼3 minutes.

3. **P3: CMA-ES polish.** Top 50 combinations polished with full $(N{+}2)$-dimensional CMA-ES (population 4096, 200K function evaluations, 3 restarts each) on a 20K-point evaluation grid. Runtime: ∼1 minute on 5 GPUs.

Total pipeline: ∼5 minutes vs ∼10 hours for pure CMA-ES screening (∼120× speedup).

A.4 Stage 4: Verification

All champion results are verified independently:

1. **150-digit verification.** Using `mpmath` with 150 decimal digits, the exact perimeter $P = 4a\,E(e^2)$ is computed at 5,000 uniformly-spaced eccentricity values and compared with the formula output. The reported $\max|\varepsilon|$ is the supremum over this grid.

2. **Exact rational tower coefficients.** All Padé tower layers use Python `Fraction` arithmetic (arbitrary-precision rationals), eliminating floating-point error in the tower construction. The exponential correction parameters are the only floating-point quantities.

3. **Cross-validation.** The R6 base error profile $\delta(h) = P_{R_6}(h)/P_{\mathrm{exact}}(h) - 1$ is independently computed by `formulas.py` and the GPU kernel, with agreement to 15 significant digits.

A.5 File manifest

File	Purpose
`src/formulas.py`	All perimeter formulas (Ramanujan I/II, tower R_2–R_6, exponential corrections)
`demo/ellipse_web_demo.py`	Interactive web demo: comparison charts, winner-band visualization, error analysis
`gpu_.../tower_gpu_kernel.py`	Tower construction (`build_tower`) and base GPU evaluation
`gpu_.../tower_gpu_kernel_v7.py`	Production GPU kernel: `TowerGPUKernelV7` (FP64 batch, `GPUCMAES`)
`r_scripts/rk_tower_quad_v10.py`	Original quad search that found R6+16-exp (0.409 ppb)
`r_scripts/rk_warmstart_v5.py`	Mac MPS add-2 campaign; explored R6+18-exp (0.358 ppb coarse)
`r_scripts/rk_warmstart_v1.py`	4-phase warm-start campaign (C1–C4)
`r_scripts/rk_varpro_screen_v1.py`	VARPRO screening, first implementation (BMM per combo)
`r_scripts/rk_varpro_screen_v2.py`	VARPRO screening with Gram matrix optimization
`r_scripts/outputs/*.json`	Machine-readable results for all runs
`active/best_results.csv`	Summary CSV of all champion results

To replicate the R6+16-exp result (pure exponential, 0.018 ppb):

```
# On Apple MPS or CUDA GPU:
cd ellipse/active/research_scripts
python3 -u rk_warmstart_campaign_v5.py --skip-c3  # runs C1+C2+C6 only
# Or for the full add-2 screen (finds the champion):
python3 -u rk_warmstart_campaign_v5.py            # ~22 min on M2 Ultra

# To evaluate just the known champion:
python3 -c "
from gpu_tower_server.tower_gpu_kernel_v7 import *
from gpu_tower_server.tower_gpu_kernel import build_tower
import torch
layers = build_tower(5)
T_k = layers[-1]['T']
k = TowerGPUKernelV7(layers, T_k,
    device='cuda:0', grid_pts=20000)
rates = torch.tensor([1/3,3/7,3/4,1,1.5,3,3.24,7.56,
    15,17,20,30,35,36,45,55],
    dtype=torch.float64, device='cuda:0')
x = torch.tensor([-2.83,9.06,0.47,-0.27,-0.45,
    -5.16,-1.54,2.45,-5.87,4.13,-1.26,
```

```
    0.73,-1.33,1.46,-0.60,17.575,2.591],
    dtype=torch.float64, device='cuda:0').unsqueeze(0)
err = k.batch_eval(rates, x, mode='mixed')
print(f'Error: {err.item()*1000:.3f} ppb')
"
```

References

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B Complete Index of Discovered Formulas

How to read these formulas

Variables. Given an ellipse with semi-major axis a and semi-minor axis b:

$$h = \left(\frac{a-b}{a+b}\right)^2 \quad (= 0 \text{ for a circle, } \to 1 \text{ as } b \to 0), \qquad x = 1 - h.$$

The rational base $R_k(h)$. Each formula uses a rational function $R_k(h)$ as its starting approximation, built by stacking Padé corrections on top of Ramanujan's second formula. Each correction is a cubic-over-quadratic rational function $N_k(h)/D_k(h)$ with exact rational coefficients (Theorem 3.1), adding 5 Taylor terms per level. Here they are explicitly (decimals shown; exact rationals in the code repository):

$$R_2(h) = 1 + \frac{3h}{10 + \sqrt{4 - 3h}} \qquad \text{(Ramanujan II)},$$

$$R_3(h) = R_2(h) + h^5 \cdot \frac{2.2889\times10^{-5} + 6.6038\times10^{-6}\,h + 1.9585\times10^{-6}\,h^2 + 5.2090\times10^{-7}\,h^3}{1 - 1.35731\,h + 0.41973\,h^2},$$

$$R_4(h) = R_3(h) + h^{10} \cdot \frac{-1.9786\times10^{-7} + 1.8641\times10^{-9}\,h + 1.2381\times10^{-7}\,h^2 + 1.0833\times10^{-7}\,h^3}{1 - 2.68000\,h + 2.01361\,h^2},$$

$$R_5(h) = R_4(h) + h^{15} \cdot \frac{-1.4724\times10^{-7} - 1.1670\times10^{-7}\,h + 5.7803\times10^{-8}\,h^2 + 2.0557\times10^{-7}\,h^3}{1 - 3.88616\,h + 4.42007\,h^2},$$

$$R_6(h) = R_5(h) + h^{20} \cdot \frac{-1.0568\times10^{-6} - 1.8281\times10^{-6}\,h - 4.9648\times10^{-7}\,h^2 + 3.3822\times10^{-6}\,h^3}{1 - 5.58931\,h + 9.32427\,h^2}.$$

Each successive correction is multiplied by h^{5k}, so it is negligible near $h = 0$ (circles) and only matters near $h = 1$ (flat ellipses). The R_k base matches the exact function $B(h) = {}_2F_1(-\frac{1}{2}, -\frac{1}{2}; 1; h)$ through order h^{5k-6}.

Endpoint constraint. At $h = 1$ (the degenerate ellipse $b = 0$), the exact perimeter is $P = 4a$, so $B(1) = 4/\pi$. Each R_k is constrained so that $R_k(1)$ yields the ratio $4q_j/p_j$ for a specific convergent p_j/q_j of π-the tower "locks on" to successively better rational approximations of π:

Base	$\pi(a+b)\,R_k(1)$ yields	Convergent of π	Error budget T_k
R_2	$4a \times 7/22$	$22/7$	$+4.0 \times 10^{-4}$
R_3	$4a \times 113/355$	$355/113$	$+8.5 \times 10^{-8}$
R_4	$4a \times 33102/103993$	$103993/33102$	-1.8×10^{-10}
R_5	$4a \times 33215/104348$	$104348/33215$	$+1.1 \times 10^{-10}$
R_6	next convergent		$\sim 10^{-11}$

The *error budget* $T_k = 1 - q_j\pi/p_j$ is how far this convergent is from π; it alternates in sign and shrinks by $\sim 10^3$ per level. The exponential correction must exactly use up this budget.

The exponential correction. The complete perimeter formula is:

$$\boxed{P = \frac{\pi(a+b)\,R_k(h)}{1 - h^q \displaystyle\sum_{i=1}^{N} c_i\, e^{-\alpha_i\, x}}, \qquad x = 1 - h, \quad \alpha_i = r_i \cdot \lambda}$$

where:

- q is the **gating exponent**: the factor h^q suppresses the correction near $h = 0$ (circles, where Ramanujan is already excellent) and activates it near $h = 1$ (extreme eccentricities). Larger q means the correction "turns on" later.
- λ is the **global decay scale**: controls how fast each exponential falls off. Larger λ means sharper, more localized corrections.

- r_i are the **rates**: rational numbers (mostly from the continued fraction of π) that set the relative decay speeds. The product $\alpha_i = r_i \cdot \lambda$ is the actual exponent coefficient in each term.

- c_i are the **coefficients**: determined by constrained least-squares once λ, q, and the rates are fixed, subject to $\sum c_i = T_k$ (the error budget).

In the equations below, $\alpha_i = r_i \cdot \lambda$ is pre-multiplied so each formula is self-contained-you can evaluate it directly without knowing r_i and λ separately. Champions (best per tower level) are in **bold**.

R3 family (R_3 base, matching $B(h)$ through h^9, error budget $T_3 = +8.49 \times 10^{-8}$)
In every R3 formula below, $R_3(h)$ stands for:

$$R_3(h) = 1 + \frac{3h}{10 + \sqrt{4-3h}} \; + \; h^5 \cdot \frac{2.289\times10^{-5} + 6.604\times10^{-6}\,h + 1.959\times10^{-6}\,h^2 + 5.209\times10^{-7}\,h^3}{1 - 1.357\,h + 0.4197\,h^2}$$

R3+4 62.147 ppb

$$P = \frac{\pi(a+b)\,R_3(h)}{1 - h^{9.9200}\left(\begin{array}{l} +\,2.847\times10^{-6}\,e^{-3.3157\,x} - 2.277\times10^{-5}\,e^{-7.7365\,x} + 1.303\times10^{-5}\,e^{-25.066\,x} \\ +\,6.978\times10^{-6}\,e^{-116.05\,x} \end{array}\right)}, \quad x = 1-h$$

R3+5 59.984 ppb

$$P = \frac{\pi(a+b)\,R_3(h)}{1 - h^{9.7942}\left(\begin{array}{l} +\,1.369\times10^{-6}\,e^{-3.3786\,x} - 1.095\times10^{-5}\,e^{-7.8833\,x} - 1.095\times10^{-5}\,e^{-9.1972\,x} \\ +\,1.384\times10^{-5}\,e^{-25.542\,x} + 6.771\times10^{-6}\,e^{-118.25\,x} \end{array}\right)}, \quad x = 1-h$$

R3+6 41.470 ppb

$$P = \frac{\pi(a+b)\,R_3(h)}{1 - h^{3.7378}\left(\begin{array}{l} +\,1.084\times10^{-6}\,e^{-5.2127\,x} - 8.669\times10^{-6}\,e^{-12.163\,x} - 8.664\times10^{-6}\,e^{-14.190\,x} \\ +\,9.366\times10^{-6}\,e^{-39.408\,x} + 4.689\times10^{-6}\,e^{-91.952\,x} + 2.279\times10^{-6}\,e^{-182.44\,x} \end{array}\right)}, \quad x = 1-h$$

R3+7 12.477 ppb

$$P = \frac{\pi(a+b)\,R_3(h)}{1 - h^{5.5097}\left(\begin{array}{l} +\,1.010\times10^{-6}\,e^{-4.2223\,x} - 8.079\times10^{-6}\,e^{-9.8521\,x} - 7.379\times10^{-6}\,e^{-11.494\,x} \\ +\,2.680\times10^{-6}\,e^{-31.921\,x} + 1.212\times10^{-5}\,e^{-74.482\,x} - 3.065\times10^{-6}\,e^{-147.78\,x} \\ +\,2.796\times10^{-6}\,e^{-236.45\,x} \end{array}\right)}, \quad x = 1-h$$

R3+8 4.943 ppb

$$P = \frac{\pi(a+b)\,R_3(h)}{1 - h^{3.9512}\left(\begin{array}{l} +\,1.415\times10^{-6}\,e^{-7.4613\,x} - 1.131\times10^{-5}\,e^{-11.606\,x} - 1.126\times10^{-5}\,e^{-17.410\,x} \\ +\,7.100\times10^{-6}\,e^{-20.311\,x} + 1.051\times10^{-5}\,e^{-56.408\,x} + 3.305\times10^{-6}\,e^{-131.62\,x} \\ -\,4.230\times10^{-7}\,e^{-261.15\,x} + 7.466\times10^{-7}\,e^{-417.83\,x} \end{array}\right)}, \quad x = 1-h$$

R3+9 2.707 ppb

$$P = \frac{\pi(a+b)\,R_3(h)}{1 - h^{3.0434}\left(\begin{array}{l} +\,1.158\times10^{-6}\,e^{-6.6313\,x} - 5.789\times10^{-6}\,e^{-10.315\,x} - 5.741\times10^{-6}\,e^{-15.473\,x} \\ -\,5.521\times10^{-6}\,e^{-18.052\,x} + 1.154\times10^{-5}\,e^{-50.133\,x} + 3.303\times10^{-6}\,e^{-116.98\,x} \\ +\,1.571\times10^{-6}\,e^{-232.10\,x} - 1.906\times10^{-6}\,e^{-371.35\,x} + 1.475\times10^{-6}\,e^{-464.19\,x} \end{array}\right)}, \quad x = 1-h$$

R3+10 2.434 ppb

$$P = \frac{\pi(a+b)\,R_3(h)}{1-h^{3.5054}\left(\begin{array}{l} +1.157\times10^{-6}\,e^{-6.3121\,x} -5.493\times10^{-6}\,e^{-9.8189\,x} -5.747\times10^{-6}\,e^{-14.728\,x} \\ -5.662\times10^{-6}\,e^{-17.183\,x} +9.683\times10^{-6}\,e^{-47.720\,x} +9.314\times10^{-6}\,e^{-95.440\,x} \\ -5.786\times10^{-6}\,e^{-111.35\,x} +4.133\times10^{-6}\,e^{-220.92\,x} -3.902\times10^{-6}\,e^{-353.48\,x} \\ +2.388\times10^{-6}\,e^{-441.85\,x} \end{array}\right)}, \quad x = 1-h$$

R3+11 2.372 ppb

$$P = \frac{\pi(a+b)\,R_3(h)}{1-h^{3.5774}\left(\begin{array}{l} +1.180\times10^{-6}\,e^{-6.3895\,x} -5.901\times10^{-6}\,e^{-9.9393\,x} \\ -5.724\times10^{-6}\,e^{-14.909\,x} -5.879\times10^{-6}\,e^{-17.394\,x} \\ +7.871\times10^{-7}\,e^{-22.363\,x} +9.588\times10^{-6}\,e^{-48.305\,x} \\ +9.346\times10^{-6}\,e^{-96.610\,x} -5.901\times10^{-6}\,e^{-112.71\,x} \\ +4.090\times10^{-6}\,e^{-223.63\,x} -3.846\times10^{-6}\,e^{-357.81\,x} \\ +2.345\times10^{-6}\,e^{-447.27\,x} \end{array}\right)}, \quad x = 1-h$$

R3+12 2.326 ppb

$$P = \frac{\pi(a+b)\,R_3(h)}{1-h^{2.7386}\left(\begin{array}{l} +1.017\times10^{-6}\,e^{-6.5886\,x} -5.011\times10^{-6}\,e^{-10.249\,x} \\ -5.074\times10^{-6}\,e^{-15.373\,x} -4.830\times10^{-6}\,e^{-17.936\,x} \\ -4.863\times10^{-6}\,e^{-23.060\,x} +4.967\times10^{-6}\,e^{-32.592\,x} \\ +7.825\times10^{-6}\,e^{-49.810\,x} +8.642\times10^{-6}\,e^{-99.619\,x} \\ -4.830\times10^{-6}\,e^{-116.22\,x} +3.482\times10^{-6}\,e^{-230.60\,x} \\ -3.319\times10^{-6}\,e^{-368.96\,x} +2.080\times10^{-6}\,e^{-461.20\,x} \end{array}\right)}, \quad x = 1-h$$

R3+13 1.835 ppb

$$P = \frac{\pi(a+b)\,R_3(h)}{1-h^{3.5796}\left(\begin{array}{l} +1.111\times10^{-6}\,e^{-6.1889\,x} -5.274\times10^{-6}\,e^{-9.6272\,x} \\ -5.274\times10^{-6}\,e^{-14.441\,x} -5.275\times10^{-6}\,e^{-16.848\,x} \\ -1.568\times10^{-6}\,e^{-21.661\,x} -1.402\times10^{-7}\,e^{-30.615\,x} \\ +1.106\times10^{-5}\,e^{-46.788\,x} +1.974\times10^{-6}\,e^{-93.577\,x} \\ +2.321\times10^{-6}\,e^{-109.17\,x} +4.405\times10^{-7}\,e^{-216.61\,x} \\ +3.000\times10^{-6}\,e^{-346.58\,x} -5.290\times10^{-6}\,e^{-433.23\,x} \\ +3.005\times10^{-6}\,e^{-505.43\,x} \end{array}\right)}, \quad x = 1-h$$

R3+14 1.495 ppb

$$P = \frac{\pi(a+b)\,R_3(h)}{1-h^{3.5796}\left(\begin{array}{l} +1.029\times10^{-6}\,e^{-6.2073\,x} -5.130\times10^{-6}\,e^{-9.6558\,x} \\ -4.889\times10^{-6}\,e^{-14.484\,x} -5.043\times10^{-6}\,e^{-16.898\,x} \\ -4.890\times10^{-6}\,e^{-21.725\,x} +5.303\times10^{-6}\,e^{-30.705\,x} \\ +6.894\times10^{-6}\,e^{-46.927\,x} +5.281\times10^{-6}\,e^{-93.854\,x} \\ +1.186\times10^{-6}\,e^{-109.50\,x} -4.891\times10^{-6}\,e^{-217.25\,x} \\ +9.320\times10^{-6}\,e^{-281.95\,x} -2.581\times10^{-6}\,e^{-347.61\,x} \\ -4.980\times10^{-6}\,e^{-434.51\,x} +3.475\times10^{-6}\,e^{-506.93\,x} \end{array}\right)}, \quad x = 1-h$$

R3+15 1.427 ppb

$$P = \frac{\pi(a{+}b)\,R_3(h)}{1-h^{3.4164}\left(\begin{array}{l} +1.069{\times}10^{-6}\,e^{-6.2414\,x} - 5.345{\times}10^{-6}\,e^{-9.7089\,x} \\ -4.028{\times}10^{-6}\,e^{-14.563\,x} - 5.070{\times}10^{-6}\,e^{-16.991\,x} \\ -5.344{\times}10^{-6}\,e^{-21.845\,x} + 3.808{\times}10^{-6}\,e^{-30.874\,x} \\ +9.540{\times}10^{-6}\,e^{-47.185\,x} - 3.477{\times}10^{-6}\,e^{-94.370\,x} \\ +1.034{\times}10^{-5}\,e^{-110.10\,x} - 3.937{\times}10^{-6}\,e^{-189.32\,x} \\ -3.353{\times}10^{-6}\,e^{-218.45\,x} + 1.059{\times}10^{-5}\,e^{-283.50\,x} \\ -3.064{\times}10^{-6}\,e^{-349.52\,x} - 5.329{\times}10^{-6}\,e^{-436.90\,x} \\ +3.678{\times}10^{-6}\,e^{-509.72\,x} \end{array}\right)}, \quad x = 1{-}h$$

R4 family (R_4 base, matching $B(h)$ through h^{14}, error budget $T_4 = -1.84 \times 10^{-10}$)
In every R4 formula below, $R_4(h)$ stands for $R_3(h)$ (defined above) plus a second Padé layer:

$$R_4(h) = R_3(h) \;+\; h^{10}\cdot\frac{-1.979{\times}10^{-7} + 1.864{\times}10^{-9}\,h + 1.238{\times}10^{-7}\,h^2 + 1.083{\times}10^{-7}\,h^3}{1 - 2.680\,h + 2.014\,h^2}$$

R4+7 3.634 ppb

$$P = \frac{\pi(a{+}b)\,R_4(h)}{1-h^{10.8247}\left(\begin{array}{l} -4.354{\times}10^{-6}\,e^{-5.8430\,x} - 1.389{\times}10^{-5}\,e^{-7.5125\,x} + 5.850{\times}10^{-6}\,e^{-17.529\,x} \\ +9.510{\times}10^{-6}\,e^{-56.794\,x} + 2.367{\times}10^{-6}\,e^{-132.52\,x} + 1.474{\times}10^{-7}\,e^{-262.94\,x} \\ +3.708{\times}10^{-7}\,e^{-525.87\,x} \end{array}\right)}, \quad x = 1{-}h$$

R4+8 3.612 ppb

$$P = \frac{\pi(a{+}b)\,R_4(h)}{1-h^{11.0642}\left(\begin{array}{l} -3.559{\times}10^{-6}\,e^{-5.5058\,x} - 1.448{\times}10^{-5}\,e^{-7.0789\,x} + 5.228{\times}10^{-6}\,e^{-16.517\,x} \\ +9.424{\times}10^{-6}\,e^{-53.516\,x} + 2.881{\times}10^{-6}\,e^{-124.87\,x} + 9.443{\times}10^{-8}\,e^{-247.76\,x} \\ +3.257{\times}10^{-7}\,e^{-495.52\,x} + 8.161{\times}10^{-8}\,e^{-578.11\,x} \end{array}\right)}, \quad x = 1{-}h$$

R4+9 2.258 ppb

$$P = \frac{\pi(a{+}b)\,R_4(h)}{1-h^{13.3435}\left(\begin{array}{l} -9.724{\times}10^{-6}\,e^{-4.4909\,x} - 2.508{\times}10^{-5}\,e^{-5.7740\,x} + 4.743{\times}10^{-5}\,e^{-10.105\,x} \\ -2.873{\times}10^{-5}\,e^{-13.473\,x} + 1.255{\times}10^{-5}\,e^{-43.652\,x} + 1.690{\times}10^{-6}\,e^{-101.85\,x} \\ +2.215{\times}10^{-6}\,e^{-202.09\,x} - 2.410{\times}10^{-6}\,e^{-404.18\,x} + 2.055{\times}10^{-6}\,e^{-471.55\,x} \end{array}\right)}, \quad x = 1{-}h$$

R4+10 1.965 ppb

$$P = \frac{\pi(a{+}b)\,R_4(h)}{1-h^{13.2997}\left(\begin{array}{l} -9.174{\times}10^{-6}\,e^{-4.5482\,x} - 2.497{\times}10^{-5}\,e^{-5.8477\,x} + 4.358{\times}10^{-5}\,e^{-10.234\,x} \\ -2.485{\times}10^{-5}\,e^{-13.645\,x} + 1.108{\times}10^{-5}\,e^{-44.209\,x} + 8.037{\times}10^{-6}\,e^{-103.15\,x} \\ -6.873{\times}10^{-6}\,e^{-122.80\,x} + 3.865{\times}10^{-6}\,e^{-204.67\,x} - 3.208{\times}10^{-6}\,e^{-409.34\,x} \\ +2.503{\times}10^{-6}\,e^{-477.56\,x} \end{array}\right)}, \quad x = 1{-}h$$

R4+11 1.539 ppb

$$P = \frac{\pi(a+b)\,R_4(h)}{1-h^{2.0246}\begin{pmatrix} -1.209\times10^{-6}\,e^{-5.0779\,x} & +2.959\times10^{-6}\,e^{-6.5287\,x} \\ -2.882\times10^{-6}\,e^{-11.425\,x} & -1.421\times10^{-5}\,e^{-15.234\,x} \\ +9.389\times10^{-6}\,e^{-49.357\,x} & +9.206\times10^{-6}\,e^{-115.17\,x} \\ +5.084\times10^{-6}\,e^{-137.10\,x} & -1.275\times10^{-5}\,e^{-150.29\,x} \\ +5.218\times10^{-6}\,e^{-228.50\,x} & -3.140\times10^{-6}\,e^{-457.01\,x} \\ +2.334\times10^{-6}\,e^{-533.18\,x} & \end{pmatrix}}, \quad x = 1-h$$

R4+12 1.136 ppb

$$P = \frac{\pi(a+b)\,R_4(h)}{1-h^{0.8519}\begin{pmatrix} -9.848\times10^{-7}\,e^{-6.0323\,x} & +2.761\times10^{-6}\,e^{-7.7558\,x} \\ -6.892\times10^{-6}\,e^{-13.573\,x} & -1.121\times10^{-5}\,e^{-18.097\,x} \\ +6.365\times10^{-6}\,e^{-42.226\,x} & +3.462\times10^{-6}\,e^{-58.634\,x} \\ +3.073\times10^{-6}\,e^{-83.877\,x} & +7.361\times10^{-6}\,e^{-136.81\,x} \\ -6.178\times10^{-6}\,e^{-162.87\,x} & +2.682\times10^{-6}\,e^{-271.45\,x} \\ -1.948\times10^{-6}\,e^{-542.90\,x} & +1.505\times10^{-6}\,e^{-633.39\,x} \end{pmatrix}}, \quad x = 1-h$$

R4+13 1.073 ppb

$$P = \frac{\pi(a+b)\,R_4(h)}{1-h^{0.8803}\begin{pmatrix} -1.061\times10^{-6}\,e^{-6.2622\,x} & +3.105\times10^{-6}\,e^{-8.0514\,x} \\ -9.878\times10^{-6}\,e^{-14.090\,x} & -8.387\times10^{-6}\,e^{-18.787\,x} \\ +6.958\times10^{-6}\,e^{-43.835\,x} & +3.713\times10^{-6}\,e^{-60.869\,x} \\ -3.628\times10^{-6}\,e^{-87.074\,x} & +7.235\times10^{-6}\,e^{-93.933\,x} \\ +3.915\times10^{-6}\,e^{-142.03\,x} & -3.867\times10^{-6}\,e^{-169.08\,x} \\ +2.276\times10^{-6}\,e^{-281.80\,x} & -1.737\times10^{-6}\,e^{-563.60\,x} \\ +1.357\times10^{-6}\,e^{-657.53\,x} & \end{pmatrix}}, \quad x = 1-h$$

R4+14 0.936 ppb

$$P = \frac{\pi(a+b)\,R_4(h)}{1-h^{1.1845}\begin{pmatrix} -1.200\times10^{-6}\,e^{-6.1369\,x} & +3.450\times10^{-6}\,e^{-7.8902\,x} \\ -1.057\times10^{-5}\,e^{-13.808\,x} & -7.698\times10^{-6}\,e^{-18.411\,x} \\ +5.824\times10^{-6}\,e^{-42.958\,x} & +4.336\times10^{-6}\,e^{-59.650\,x} \\ +3.679\times10^{-6}\,e^{-85.331\,x} & -7.430\times10^{-7}\,e^{-92.053\,x} \\ +1.765\times10^{-6}\,e^{-139.18\,x} & +1.599\times10^{-6}\,e^{-165.70\,x} \\ -8.321\times10^{-6}\,e^{-276.16\,x} & +9.117\times10^{-6}\,e^{-312.98\,x} \\ -3.454\times10^{-6}\,e^{-552.32\,x} & +2.219\times10^{-6}\,e^{-644.37\,x} \end{pmatrix}}, \quad x = 1-h$$

R4+15 0.786 ppb

$$P = \frac{\pi(a+b)\,R_4(h)}{1-h^{1.4531}\begin{pmatrix} -1.341\times10^{-6}\,e^{-6.1633\,x} & +3.893\times10^{-6}\,e^{-7.9242\,x} \\ -1.289\times10^{-5}\,e^{-13.867\,x} & -5.414\times10^{-6}\,e^{-18.490\,x} \\ +5.405\times10^{-6}\,e^{-43.143\,x} & +3.963\times10^{-6}\,e^{-59.907\,x} \\ +7.588\times10^{-6}\,e^{-85.699\,x} & -2.905\times10^{-6}\,e^{-92.449\,x} \\ -6.271\times10^{-6}\,e^{-139.78\,x} & +1.069\times10^{-5}\,e^{-166.41\,x} \\ -1.300\times10^{-5}\,e^{-277.35\,x} & +1.016\times10^{-5}\,e^{-314.33\,x} \\ +4.677\times10^{-6}\,e^{-462.24\,x} & -8.371\times10^{-6}\,e^{-554.69\,x} \\ +3.817\times10^{-6}\,e^{-647.14\,x} & \end{pmatrix}}, \quad x = 1-h$$

R4+16 0.741 ppb

$$P = \frac{\pi(a+b)\,R_4(h)}{1-h^{1.5901}\begin{pmatrix} -1.372\times10^{-6}\,e^{-6.0374\,x} & +3.917\times10^{-6}\,e^{-7.7624\,x} \\ -1.210\times10^{-5}\,e^{-13.584\,x} & -6.306\times10^{-6}\,e^{-18.112\,x} \\ +5.672\times10^{-6}\,e^{-42.262\,x} & +2.917\times10^{-6}\,e^{-58.684\,x} \\ +7.734\times10^{-6}\,e^{-83.949\,x} & -1.584\times10^{-6}\,e^{-90.561\,x} \\ -7.266\times10^{-6}\,e^{-136.93\,x} & +1.064\times10^{-5}\,e^{-163.01\,x} \\ -9.803\times10^{-6}\,e^{-271.68\,x} & +6.820\times10^{-6}\,e^{-307.91\,x} \\ +4.884\times10^{-6}\,e^{-452.81\,x} & -6.771\times10^{-6}\,e^{-543.37\,x} \\ +2.338\times10^{-6}\,e^{-633.93\,x} & +2.764\times10^{-7}\,e^{-815.05\,x} \end{pmatrix}}, \quad x = 1-h$$

R4+17 0.809 ppb

$$P = \frac{\pi(a+b)\,R_4(h)}{1-h^{2.3358}\begin{pmatrix} -1.471\times10^{-6}\,e^{-5.5986\,x} & +4.091\times10^{-6}\,e^{-7.1982\,x} \\ -1.081\times10^{-5}\,e^{-12.597\,x} & -7.696\times10^{-6}\,e^{-16.796\,x} \\ +4.686\times10^{-6}\,e^{-39.190\,x} & +2.243\times10^{-6}\,e^{-50.387\,x} \\ -3.188\times10^{-7}\,e^{-54.418\,x} & +1.031\times10^{-5}\,e^{-77.847\,x} \\ -1.669\times10^{-6}\,e^{-83.979\,x} & -8.981\times10^{-6}\,e^{-126.98\,x} \\ +1.136\times10^{-5}\,e^{-151.16\,x} & -4.717\times10^{-6}\,e^{-251.94\,x} \\ +5.001\times10^{-7}\,e^{-285.53\,x} & +8.608\times10^{-6}\,e^{-419.89\,x} \\ -9.109\times10^{-6}\,e^{-503.87\,x} & +2.522\times10^{-6}\,e^{-587.85\,x} \\ +4.492\times10^{-7}\,e^{-755.81\,x} & \end{pmatrix}}, \quad x = 1-h$$

R4+18 0.685 ppb

$$P = \frac{\pi(a+b)\,R_4(h)}{1-h^{3.3367}\begin{pmatrix} -1.542\times10^{-6}\,e^{-5.5843\,x} & +4.594\times10^{-6}\,e^{-7.1798\,x} \\ -1.469\times10^{-5}\,e^{-12.565\,x} & -1.110\times10^{-5}\,e^{-16.753\,x} \\ +9.178\times10^{-6}\,e^{-19.545\,x} & -3.572\times10^{-6}\,e^{-39.090\,x} \\ +9.883\times10^{-6}\,e^{-50.259\,x} & +1.463\times10^{-6}\,e^{-54.279\,x} \\ +2.151\times10^{-6}\,e^{-77.648\,x} & +2.618\times10^{-6}\,e^{-83.764\,x} \\ -8.783\times10^{-6}\,e^{-126.65\,x} & +1.147\times10^{-5}\,e^{-150.78\,x} \\ -1.930\times10^{-6}\,e^{-251.29\,x} & -2.860\times10^{-6}\,e^{-284.80\,x} \\ +7.540\times10^{-6}\,e^{-418.82\,x} & -4.332\times10^{-6}\,e^{-502.59\,x} \\ -1.133\times10^{-6}\,e^{-586.35\,x} & +1.043\times10^{-6}\,e^{-753.88\,x} \end{pmatrix}}, \quad x = 1-h$$

R4+19 0.525 ppb

$$P = \frac{\pi(a{+}b)\,R_4(h)}{1-h^{2.0604}\left(\begin{array}{ll} -1.324{\times}10^{-6}\,e^{-6.0306\,x} & +3.832{\times}10^{-6}\,e^{-7.7536\,x} \\ -1.175{\times}10^{-5}\,e^{-13.569\,x} & -1.465{\times}10^{-5}\,e^{-18.092\,x} \\ +1.004{\times}10^{-5}\,e^{-21.107\,x} & -1.662{\times}10^{-6}\,e^{-42.214\,x} \\ +8.143{\times}10^{-6}\,e^{-54.275\,x} & +2.946{\times}10^{-6}\,e^{-58.617\,x} \\ -2.490{\times}10^{-6}\,e^{-83.854\,x} & +5.923{\times}10^{-6}\,e^{-90.459\,x} \\ -5.525{\times}10^{-6}\,e^{-136.77\,x} & +7.013{\times}10^{-6}\,e^{-162.83\,x} \\ +1.086{\times}10^{-6}\,e^{-271.38\,x} & -3.777{\times}10^{-6}\,e^{-307.56\,x} \\ +4.737{\times}10^{-6}\,e^{-452.29\,x} & -3.289{\times}10^{-6}\,e^{-542.75\,x} \\ +7.622{\times}10^{-7}\,e^{-633.21\,x} & -1.405{\times}10^{-7}\,e^{-814.13\,x} \\ +1.216{\times}10^{-7}\,e^{-1266.4\,x} & \end{array}\right)}, \quad x = 1{-}h$$

R4+20 0.492 ppb

$$P = \frac{\pi(a{+}b)\,R_4(h)}{1-h^{2.5199}\left(\begin{array}{ll} -1.594{\times}10^{-6}\,e^{-5.2674\,x} & +4.005{\times}10^{-6}\,e^{-6.7724\,x} \\ -6.078{\times}10^{-6}\,e^{-11.852\,x} & -1.789{\times}10^{-5}\,e^{-15.802\,x} \\ +7.005{\times}10^{-6}\,e^{-18.436\,x} & -1.534{\times}10^{-6}\,e^{-36.872\,x} \\ +8.604{\times}10^{-6}\,e^{-47.407\,x} & -5.919{\times}10^{-7}\,e^{-51.199\,x} \\ +4.016{\times}10^{-6}\,e^{-73.242\,x} & +2.309{\times}10^{-6}\,e^{-79.011\,x} \\ -4.071{\times}10^{-6}\,e^{-119.47\,x} & +6.387{\times}10^{-6}\,e^{-142.22\,x} \\ -4.451{\times}10^{-7}\,e^{-189.63\,x} & -7.202{\times}10^{-6}\,e^{-237.03\,x} \\ +9.244{\times}10^{-6}\,e^{-268.64\,x} & -3.986{\times}10^{-6}\,e^{-395.06\,x} \\ -1.640{\times}10^{-6}\,e^{-474.07\,x} & +5.453{\times}10^{-6}\,e^{-553.08\,x} \\ -2.320{\times}10^{-6}\,e^{-711.10\,x} & +3.261{\times}10^{-7}\,e^{-1106.2\,x} \end{array}\right)}, \quad x = 1{-}h$$

R5 family (R_5 base, matching $B(h)$ through h^{19}, error budget $T_5 = +1.06 \times 10^{-10}$)
In every R5 formula below, $R_5(h)$ stands for $R_4(h)$ (defined above) plus a third Padé layer:

$$R_5(h) = R_4(h) \;+\; h^{15}\cdot\frac{-1.472{\times}10^{-7} - 1.167{\times}10^{-7}\,h + 5.780{\times}10^{-8}\,h^2 + 2.056{\times}10^{-7}\,h^3}{1 - 3.886\,h + 4.420\,h^2}$$

R5+8 3.728 ppb

$$P = \frac{\pi(a{+}b)\,R_5(h)}{1-h^{3.3124}\left(\begin{array}{lll} +2.001{\times}10^{-6}\,e^{-8.7561\,x} & -1.549{\times}10^{-5}\,e^{-13.621\,x} & -1.600{\times}10^{-5}\,e^{-20.431\,x} \\ +1.749{\times}10^{-5}\,e^{-23.836\,x} & +9.425{\times}10^{-6}\,e^{-66.196\,x} & +2.289{\times}10^{-6}\,e^{-154.46\,x} \\ -2.308{\times}10^{-7}\,e^{-306.47\,x} & +5.228{\times}10^{-7}\,e^{-490.34\,x} & \end{array}\right)}, \quad x = 1{-}h$$

R5+9 1.523 ppb

$$P = \frac{\pi(a{+}b)\,R_5(h)}{1-h^{2.9717}\left(\begin{array}{lll} +1.968{\times}10^{-6}\,e^{-8.8355\,x} & -1.495{\times}10^{-5}\,e^{-13.744\,x} & -1.502{\times}10^{-5}\,e^{-20.616\,x} \\ +1.567{\times}10^{-5}\,e^{-24.052\,x} & +1.023{\times}10^{-5}\,e^{-66.797\,x} & +1.055{\times}10^{-6}\,e^{-155.86\,x} \\ +2.044{\times}10^{-6}\,e^{-309.24\,x} & -1.075{\times}10^{-5}\,e^{-494.79\,x} & +9.762{\times}10^{-6}\,e^{-515.41\,x} \end{array}\right)}, \quad x = 1{-}h$$

R5+10 1.451 ppb

$$P = \frac{\pi(a+b)\,R_5(h)}{1-h^{2.6310}\left(\begin{array}{l} +1.862\times10^{-6}\,e^{-9.0571\,x} -1.473\times10^{-5}\,e^{-14.089\,x} -1.439\times10^{-5}\,e^{-21.133\,x} \\ +1.398\times10^{-5}\,e^{-24.656\,x} +1.324\times10^{-6}\,e^{-31.700\,x} +9.925\times10^{-6}\,e^{-68.472\,x} \\ +1.030\times10^{-6}\,e^{-159.77\,x} +1.935\times10^{-6}\,e^{-317.00\,x} -1.020\times10^{-5}\,e^{-507.20\,x} \\ +9.260\times10^{-6}\,e^{-528.33\,x} \end{array}\right)}, \quad x = 1-h$$

R5+11 1.221 ppb

$$P = \frac{\pi(a+b)\,R_5(h)}{1-h^{3.3831}\left(\begin{array}{l} +2.088\times10^{-6}\,e^{-8.4496\,x} -1.445\times10^{-5}\,e^{-13.144\,x} \\ -1.600\times10^{-5}\,e^{-19.716\,x} +1.612\times10^{-5}\,e^{-23.002\,x} \\ -5.104\times10^{-7}\,e^{-29.574\,x} +1.028\times10^{-5}\,e^{-63.879\,x} \\ +1.708\times10^{-6}\,e^{-149.05\,x} +5.312\times10^{-7}\,e^{-295.74\,x} \\ +1.329\times10^{-5}\,e^{-473.18\,x} -1.587\times10^{-5}\,e^{-492.89\,x} \\ +2.824\times10^{-6}\,e^{-591.47\,x} \end{array}\right)}, \quad x = 1-h$$

R5+12 1.189 ppb

$$P = \frac{\pi(a+b)\,R_5(h)}{1-h^{3.1473}\left(\begin{array}{l} +2.069\times10^{-6}\,e^{-8.8213\,x} -1.593\times10^{-5}\,e^{-13.722\,x} \\ -1.444\times10^{-5}\,e^{-20.583\,x} +1.745\times10^{-5}\,e^{-24.014\,x} \\ -1.916\times10^{-6}\,e^{-30.875\,x} +1.162\times10^{-5}\,e^{-66.689\,x} \\ -2.060\times10^{-6}\,e^{-102.92\,x} +2.795\times10^{-6}\,e^{-155.61\,x} \\ -7.581\times10^{-8}\,e^{-308.75\,x} +1.365\times10^{-5}\,e^{-494.00\,x} \\ -1.573\times10^{-5}\,e^{-514.58\,x} +2.567\times10^{-6}\,e^{-617.49\,x} \end{array}\right)}, \quad x = 1-h$$

R5+13 0.854 ppb

$$P = \frac{\pi(a+b)\,R_5(h)}{1-h^{4.1502}\left(\begin{array}{l} +2.514\times10^{-6}\,e^{-8.2603\,x} -1.758\times10^{-5}\,e^{-12.849\,x} \\ -1.797\times10^{-5}\,e^{-19.274\,x} +2.854\times10^{-5}\,e^{-22.486\,x} \\ -1.034\times10^{-5}\,e^{-28.911\,x} +1.619\times10^{-5}\,e^{-62.448\,x} \\ -8.493\times10^{-6}\,e^{-96.371\,x} +9.155\times10^{-6}\,e^{-145.71\,x} \\ -1.124\times10^{-5}\,e^{-289.11\,x} +2.084\times10^{-5}\,e^{-385.48\,x} \\ +1.272\times10^{-6}\,e^{-462.58\,x} -1.905\times10^{-5}\,e^{-481.85\,x} \\ +6.166\times10^{-6}\,e^{-578.22\,x} \end{array}\right)}, \quad x = 1-h$$

R5+14 0.812 ppb

$$P = \frac{\pi(a+b)\,R_5(h)}{1-h^{4.0554}\left(\begin{array}{l} +2.587\times10^{-6}\,e^{-8.5399\,x} -1.988\times10^{-5}\,e^{-13.284\,x} \\ -1.230\times10^{-5}\,e^{-19.926\,x} +2.560\times10^{-5}\,e^{-23.247\,x} \\ -1.112\times10^{-5}\,e^{-29.890\,x} +1.894\times10^{-5}\,e^{-64.562\,x} \\ -1.639\times10^{-5}\,e^{-99.632\,x} +2.778\times10^{-5}\,e^{-139.48\,x} \\ -1.431\times10^{-5}\,e^{-150.64\,x} -8.047\times10^{-6}\,e^{-298.90\,x} \\ +1.629\times10^{-5}\,e^{-398.53\,x} +4.675\times10^{-6}\,e^{-478.23\,x} \\ -1.943\times10^{-5}\,e^{-498.16\,x} +5.600\times10^{-6}\,e^{-597.79\,x} \end{array}\right)}, \quad x = 1-h$$

R5+15 0.821 ppb

$$P = \frac{\pi(a+b)\,R_5(h)}{1-h^{4.1407}\left(\begin{array}{ll} +2.547\times10^{-6}\,e^{-8.3605\,x} & -1.846\times10^{-5}\,e^{-13.005\,x} \\ -1.669\times10^{-5}\,e^{-19.508\,x} & +2.894\times10^{-5}\,e^{-22.759\,x} \\ -1.136\times10^{-5}\,e^{-29.262\,x} & +1.722\times10^{-5}\,e^{-63.206\,x} \\ -1.096\times10^{-5}\,e^{-97.540\,x} & +1.453\times10^{-5}\,e^{-136.56\,x} \\ -7.378\times10^{-6}\,e^{-147.48\,x} & +5.407\times10^{-6}\,e^{-195.08\,x} \\ -1.493\times10^{-5}\,e^{-292.62\,x} & +2.437\times10^{-5}\,e^{-390.16\,x} \\ -5.148\times10^{-7}\,e^{-468.19\,x} & -1.915\times10^{-5}\,e^{-487.70\,x} \\ +6.436\times10^{-6}\,e^{-585.24\,x} & \end{array}\right)}, \quad x = 1-h$$

R5+16 0.599 ppb

$$P = \frac{\pi(a+b)\,R_5(h)}{1-h^{6.5194}\left(\begin{array}{ll} +3.860\times10^{-6}\,e^{-6.1489\,x} & -1.547\times10^{-5}\,e^{-9.5649\,x} \\ -3.005\times10^{-5}\,e^{-14.347\,x} & +4.072\times10^{-5}\,e^{-16.739\,x} \\ -1.482\times10^{-5}\,e^{-21.521\,x} & +7.743\times10^{-6}\,e^{-46.486\,x} \\ +1.019\times10^{-5}\,e^{-71.737\,x} & -1.606\times10^{-6}\,e^{-100.43\,x} \\ -1.528\times10^{-5}\,e^{-108.47\,x} & +2.405\times10^{-5}\,e^{-143.47\,x} \\ -2.349\times10^{-5}\,e^{-215.21\,x} & +2.685\times10^{-5}\,e^{-286.95\,x} \\ -7.344\times10^{-6}\,e^{-344.34\,x} & -9.678\times10^{-6}\,e^{-358.69\,x} \\ +4.265\times10^{-6}\,e^{-430.42\,x} & +4.337\times10^{-8}\,e^{-1434.7\,x} \end{array}\right)}, \quad x = 1-h$$

R5+17 0.649 ppb

$$P = \frac{\pi(a+b)\,R_5(h)}{1-h^{5.8454}\left(\begin{array}{ll} +3.440\times10^{-6}\,e^{-6.8177\,x} & -1.677\times10^{-5}\,e^{-10.605\,x} \\ -2.638\times10^{-5}\,e^{-15.908\,x} & +3.921\times10^{-5}\,e^{-18.559\,x} \\ -1.558\times10^{-5}\,e^{-23.862\,x} & +1.280\times10^{-5}\,e^{-51.542\,x} \\ -1.311\times10^{-6}\,e^{-79.540\,x} & +2.072\times10^{-5}\,e^{-111.36\,x} \\ -2.611\times10^{-5}\,e^{-120.26\,x} & +1.730\times10^{-5}\,e^{-159.08\,x} \\ -1.022\times10^{-5}\,e^{-190.90\,x} & +6.087\times10^{-6}\,e^{-238.62\,x} \\ -1.025\times10^{-5}\,e^{-318.16\,x} & -2.077\times10^{-5}\,e^{-381.79\,x} \\ +3.767\times10^{-5}\,e^{-397.70\,x} & -1.052\times10^{-5}\,e^{-477.24\,x} \\ +6.927\times10^{-7}\,e^{-795.40\,x} & \end{array}\right)}, \quad x = 1-h$$

R5+18 0.573 ppb

$$P = \frac{\pi(a+b)\,R_5(h)}{1-h^{6.2305}\left(\begin{array}{ll} +3.872\times10^{-6}\,e^{-6.6360\,x} & -1.844\times10^{-5}\,e^{-10.323\,x} \\ -2.657\times10^{-5}\,e^{-15.484\,x} & +4.331\times10^{-5}\,e^{-18.065\,x} \\ -1.881\times10^{-5}\,e^{-23.226\,x} & +1.316\times10^{-5}\,e^{-50.168\,x} \\ -3.421\times10^{-6}\,e^{-71.767\,x} & +5.715\times10^{-6}\,e^{-77.420\,x} \\ +1.811\times10^{-6}\,e^{-108.39\,x} & -1.161\times10^{-5}\,e^{-117.06\,x} \\ +1.929\times10^{-5}\,e^{-154.84\,x} & -2.178\times10^{-5}\,e^{-232.26\,x} \\ +2.408\times10^{-5}\,e^{-309.68\,x} & +7.919\times10^{-6}\,e^{-371.61\,x} \\ -2.293\times10^{-5}\,e^{-387.10\,x} & -1.289\times10^{-6}\,e^{-402.58\,x} \\ +5.660\times10^{-6}\,e^{-464.52\,x} & +3.327\times10^{-8}\,e^{-1548.4\,x} \end{array}\right)}, \quad x = 1-h$$

R5+19 0.571 ppb

$$P = \frac{\pi(a{+}b)\,R_5(h)}{1-h^{6.2123}\left(\begin{array}{ll} +3.918{\times}10^{-6}\,e^{-6.6892\,x} & -1.910{\times}10^{-5}\,e^{-10.405\,x} \\ -2.503{\times}10^{-5}\,e^{-15.608\,x} & +4.269{\times}10^{-5}\,e^{-18.210\,x} \\ -1.938{\times}10^{-5}\,e^{-23.412\,x} & +1.537{\times}10^{-5}\,e^{-50.571\,x} \\ -1.763{\times}10^{-5}\,e^{-72.343\,x} & +2.011{\times}10^{-5}\,e^{-78.041\,x} \\ -1.073{\times}10^{-5}\,e^{-109.26\,x} & +1.427{\times}10^{-6}\,e^{-118.00\,x} \\ +2.569{\times}10^{-5}\,e^{-156.08\,x} & -1.601{\times}10^{-5}\,e^{-171.69\,x} \\ -7.689{\times}10^{-6}\,e^{-234.12\,x} & +1.031{\times}10^{-5}\,e^{-312.16\,x} \\ +1.625{\times}10^{-5}\,e^{-374.60\,x} & -2.418{\times}10^{-5}\,e^{-390.20\,x} \\ -2.234{\times}10^{-7}\,e^{-405.81\,x} & +4.180{\times}10^{-6}\,e^{-468.25\,x} \\ +3.372{\times}10^{-8}\,e^{-1560.8\,x} & \end{array}\right)}, \quad x = 1{-}h$$

R5+20 0.508 ppb

$$P = \frac{\pi(a{+}b)\,R_5(h)}{1-h^{6.6769}\left(\begin{array}{ll} +4.826{\times}10^{-6}\,e^{-6.5273\,x} & -2.400{\times}10^{-5}\,e^{-10.154\,x} \\ -1.398{\times}10^{-5}\,e^{-15.230\,x} & +3.611{\times}10^{-5}\,e^{-17.769\,x} \\ -2.063{\times}10^{-5}\,e^{-22.845\,x} & +1.836{\times}10^{-5}\,e^{-49.346\,x} \\ -3.036{\times}10^{-5}\,e^{-70.591\,x} & +2.987{\times}10^{-5}\,e^{-76.152\,x} \\ +6.103{\times}10^{-6}\,e^{-106.61\,x} & -2.172{\times}10^{-5}\,e^{-115.14\,x} \\ +4.819{\times}10^{-5}\,e^{-152.30\,x} & -3.003{\times}10^{-5}\,e^{-167.53\,x} \\ -1.871{\times}10^{-5}\,e^{-228.45\,x} & +3.540{\times}10^{-5}\,e^{-304.61\,x} \\ +2.202{\times}10^{-5}\,e^{-365.53\,x} & -3.665{\times}10^{-5}\,e^{-380.76\,x} \\ -2.688{\times}10^{-5}\,e^{-395.99\,x} & +2.303{\times}10^{-5}\,e^{-456.91\,x} \\ -1.018{\times}10^{-6}\,e^{-761.52\,x} & +9.109{\times}10^{-8}\,e^{-1523.0\,x} \end{array}\right)}, \quad x = 1{-}h$$

R6 family (R_6 base, matching $B(h)$ through h^{24}, error budget $T_6 \approx 1.7 \times 10^{-11}$)
In every R6 formula below, $R_6(h)$ stands for $R_5(h)$ (defined above) plus a fourth and final Padé layer:

$$R_6(h) = R_5(h) \;+\; h^{20}\cdot\frac{-1.057{\times}10^{-6} - 1.828{\times}10^{-6}\,h - 4.965{\times}10^{-7}\,h^2 + 3.382{\times}10^{-6}\,h^3}{1 - 5.589\,h + 9.324\,h^2}$$

The full chain is: $R_6(h) = 1 + 3h/(10{+}\sqrt{4{-}3h}) + h^5\,\psi_3(h) + h^{10}\,\psi_4(h) + h^{15}\,\psi_5(h) + h^{20}\,\psi_6(h)$, where each ψ_k is the cubic/quadratic fraction shown in the corresponding family header above.
R6+7 4.071 ppb

$$P = \frac{\pi(a{+}b)\,R_6(h)}{1-h^{1.3359}\left(\begin{array}{lll} -1.412{\times}10^{-6}\,e^{-5.1725\,x} & +2.869{\times}10^{-6}\,e^{-6.6503\,x} & -1.693{\times}10^{-5}\,e^{-15.517\,x} \\ +1.019{\times}10^{-5}\,e^{-50.276\,x} & +4.912{\times}10^{-6}\,e^{-117.31\,x} & -1.850{\times}10^{-7}\,e^{-232.76\,x} \\ +5.639{\times}10^{-7}\,e^{-465.52\,x} & & \end{array}\right)}, \quad x = 1{-}h$$

R6+8 3.613 ppb

$$P = \frac{\pi(a{+}b)\,R_6(h)}{1-h^{2.7310}\left(\begin{array}{lll} -1.458{\times}10^{-6}\,e^{-5.7547\,x} & +4.788{\times}10^{-6}\,e^{-7.3989\,x} & -1.738{\times}10^{-5}\,e^{-12.948\,x} \\ -4.965{\times}10^{-7}\,e^{-17.264\,x} & +1.074{\times}10^{-5}\,e^{-55.936\,x} & +3.345{\times}10^{-6}\,e^{-130.52\,x} \\ +4.581{\times}10^{-8}\,e^{-258.96\,x} & +4.174{\times}10^{-7}\,e^{-517.92\,x} & \end{array}\right)}, \quad x = 1{-}h$$

R6+9 1.649 ppb

$$P = \frac{\pi(a+b)\,R_6(h)}{1-h^{2.7423}\left(\begin{array}{l} -1.425\times10^{-6}\,e^{-5.8121\,x} + 4.524\times10^{-6}\,e^{-7.4727\,x} - 1.696\times10^{-5}\,e^{-13.077\,x} \\ -8.135\times10^{-7}\,e^{-17.436\,x} + 1.129\times10^{-5}\,e^{-56.493\,x} + 2.462\times10^{-6}\,e^{-131.82\,x} \\ +9.718\times10^{-7}\,e^{-261.54\,x} - 1.020\times10^{-6}\,e^{-523.09\,x} + 9.766\times10^{-7}\,e^{-610.27\,x} \end{array}\right)}, \quad x = 1-h$$

R6+10 1.332 ppb

$$P = \frac{\pi(a+b)\,R_6(h)}{1-h^{2.4825}\left(\begin{array}{l} -1.589\times10^{-6}\,e^{-5.5515\,x} + 4.498\times10^{-6}\,e^{-7.1377\,x} - 1.226\times10^{-5}\,e^{-12.491\,x} \\ -5.702\times10^{-6}\,e^{-16.655\,x} + 1.099\times10^{-5}\,e^{-53.961\,x} + 3.682\times10^{-6}\,e^{-125.91\,x} \\ -4.290\times10^{-6}\,e^{-249.82\,x} + 5.505\times10^{-6}\,e^{-283.13\,x} - 2.812\times10^{-6}\,e^{-499.64\,x} \\ +1.984\times10^{-6}\,e^{-582.91\,x} \end{array}\right)}, \quad x = 1-h$$

R6+11 0.955 ppb

$$P = \frac{\pi(a+b)\,R_6(h)}{1-h^{1.6227}\left(\begin{array}{l} -1.659\times10^{-6}\,e^{-6.2671\,x} + 4.777\times10^{-6}\,e^{-8.0577\,x} \\ -1.656\times10^{-5}\,e^{-14.101\,x} - 1.437\times10^{-6}\,e^{-18.801\,x} \\ +1.261\times10^{-5}\,e^{-56.404\,x} - 1.508\times10^{-6}\,e^{-60.916\,x} \\ +3.987\times10^{-6}\,e^{-142.14\,x} - 6.721\times10^{-6}\,e^{-282.02\,x} \\ +7.531\times10^{-6}\,e^{-319.62\,x} - 2.959\times10^{-6}\,e^{-564.04\,x} \\ +1.940\times10^{-6}\,e^{-658.04\,x} \end{array}\right)}, \quad x = 1-h$$

R6+12 0.830 ppb

$$P = \frac{\pi(a+b)\,R_6(h)}{1-h^{0.8837}\left(\begin{array}{l} -9.690\times10^{-7}\,e^{-6.5416\,x} + 2.961\times10^{-6}\,e^{-8.4106\,x} \\ -1.104\times10^{-5}\,e^{-14.719\,x} - 9.051\times10^{-6}\,e^{-19.625\,x} \\ +4.899\times10^{-6}\,e^{-29.437\,x} + 3.787\times10^{-6}\,e^{-58.874\,x} \\ +6.102\times10^{-6}\,e^{-63.584\,x} + 3.459\times10^{-6}\,e^{-148.36\,x} \\ -6.487\times10^{-6}\,e^{-294.37\,x} + 7.364\times10^{-6}\,e^{-333.62\,x} \\ -2.929\times10^{-6}\,e^{-588.74\,x} + 1.898\times10^{-6}\,e^{-686.86\,x} \end{array}\right)}, \quad x = 1-h$$

R6+13 0.623 ppb

$$P = \frac{\pi(a+b)\,R_6(h)}{1-h^{1.6030}\left(\begin{array}{l} -1.302\times10^{-6}\,e^{-6.2193\,x} + 3.826\times10^{-6}\,e^{-7.9963\,x} \\ -1.306\times10^{-5}\,e^{-13.994\,x} - 6.283\times10^{-6}\,e^{-18.658\,x} \\ +2.874\times10^{-6}\,e^{-27.987\,x} + 4.675\times10^{-6}\,e^{-55.974\,x} \\ +5.688\times10^{-6}\,e^{-60.452\,x} + 3.542\times10^{-6}\,e^{-141.05\,x} \\ -4.388\times10^{-6}\,e^{-279.87\,x} + 4.971\times10^{-6}\,e^{-317.19\,x} \\ -1.683\times10^{-6}\,e^{-559.74\,x} + 1.128\times10^{-6}\,e^{-653.03\,x} \\ +1.144\times10^{-8}\,e^{-2239.0\,x} \end{array}\right)}, \quad x = 1-h$$

R6+14 0.700 ppb

$$P = \frac{\pi(a+b)\,R_6(h)}{1-h^{1.2615}\begin{pmatrix} -1.126\times10^{-6}\,e^{-6.4095\,x} & +3.395\times10^{-6}\,e^{-8.2408\,x} \\ -1.245\times10^{-5}\,e^{-14.421\,x} & -7.250\times10^{-6}\,e^{-19.228\,x} \\ +3.898\times10^{-6}\,e^{-28.843\,x} & +4.499\times10^{-6}\,e^{-57.685\,x} \\ +5.605\times10^{-6}\,e^{-62.300\,x} & +3.566\times10^{-6}\,e^{-145.37\,x} \\ -6.960\times10^{-6}\,e^{-288.43\,x} & +8.273\times10^{-6}\,e^{-326.88\,x} \\ -2.643\times10^{-6}\,e^{-499.94\,x} & +6.588\times10^{-7}\,e^{-576.85\,x} \\ +4.997\times10^{-7}\,e^{-673.00\,x} & +3.101\times10^{-8}\,e^{-1346.0\,x} \end{pmatrix}}, \quad x = 1-h$$

R6+15 0.477 ppb

$$P = \frac{\pi(a+b)\,R_6(h)}{1-h^{1.9386}\begin{pmatrix} -1.384\times10^{-6}\,e^{-6.1721\,x} & +4.062\times10^{-6}\,e^{-7.9356\,x} \\ -1.392\times10^{-5}\,e^{-13.887\,x} & -6.884\times10^{-6}\,e^{-18.516\,x} \\ +1.023\times10^{-5}\,e^{-27.775\,x} & -7.382\times10^{-6}\,e^{-32.404\,x} \\ +7.741\times10^{-6}\,e^{-55.549\,x} & +4.207\times10^{-6}\,e^{-59.993\,x} \\ +2.698\times10^{-6}\,e^{-139.98\,x} & +8.206\times10^{-7}\,e^{-277.75\,x} \\ -2.021\times10^{-7}\,e^{-314.78\,x} & -3.481\times10^{-6}\,e^{-481.43\,x} \\ +6.160\times10^{-6}\,e^{-555.49\,x} & -2.799\times10^{-6}\,e^{-648.07\,x} \\ +1.345\times10^{-7}\,e^{-1296.1\,x} & \end{pmatrix}}, \quad x = 1-h$$

R6+16 0.018 ppb — (this work, superseded by SCOR)

$$P = \frac{\pi(a+b)\,R_6(h)}{1-h^{7.6874}\begin{pmatrix} -7.477\times10^{-5}\,e^{-6.4814\,x} & +8.480\times10^{-4}\,e^{-8.8090\,x} \\ -1.375\times10^{-3}\,e^{-10.180\,x} & +2.151\times10^{-3}\,e^{-14.569\,x} \\ -1.958\times10^{-3}\,e^{-16.657\,x} & +5.984\times10^{-4}\,e^{-22.580\,x} \\ -2.023\times10^{-4}\,e^{-26.260\,x} & +8.698\times10^{-6}\,e^{-57.325\,x} \\ +2.452\times10^{-6}\,e^{-91.868\,x} & +1.741\times10^{-6}\,e^{-200.09\,x} \\ -5.234\times10^{-6}\,e^{-372.92\,x} & +5.453\times10^{-6}\,e^{-395.77\,x} \\ -8.113\times10^{-7}\,e^{-694.96\,x} & +5.347\times10^{-7}\,e^{-832.67\,x} \\ -2.118\times10^{-8}\,e^{-2816.7\,x} & +2.201\times10^{-8}\,e^{-3499.8\,x} \end{pmatrix}}, \quad x = 1-h$$

R6+17 0.521 ppb

$$P = \frac{\pi(a+b)\,R_6(h)}{1-h^{1.9074}\begin{pmatrix} -1.461\times10^{-6}\,e^{-6.3141\,x} & +4.393\times10^{-6}\,e^{-8.1181\,x} \\ -1.707\times10^{-5}\,e^{-14.207\,x} & -1.709\times10^{-6}\,e^{-18.942\,x} \\ +1.978\times10^{-6}\,e^{-28.414\,x} & +2.953\times10^{-6}\,e^{-56.827\,x} \\ +8.046\times10^{-6}\,e^{-61.373\,x} & -4.872\times10^{-6}\,e^{-122.75\,x} \\ +1.099\times10^{-5}\,e^{-143.20\,x} & -1.305\times10^{-5}\,e^{-208.37\,x} \\ +4.123\times10^{-6}\,e^{-246.25\,x} & +9.425\times10^{-6}\,e^{-263.39\,x} \\ +7.343\times10^{-6}\,e^{-284.14\,x} & -1.353\times10^{-5}\,e^{-322.02\,x} \\ +8.415\times10^{-6}\,e^{-568.27\,x} & -7.552\times10^{-6}\,e^{-662.98\,x} \\ +1.582\times10^{-6}\,e^{-852.41\,x} & \end{pmatrix}}, \quad x = 1-h$$

R6+18 0.358 ppb (coarse grid; true peak 115 ppb at $e > 0.999$)

$$P = \frac{\pi(a+b)\,R_6(h)}{1-h^{2.6332}\begin{pmatrix} -1.889\times10^{-6}\,e^{-5.8138\,x} & +5.261\times10^{-6}\,e^{-7.4749\,x} \\ -1.695\times10^{-5}\,e^{-13.081\,x} & -1.142\times10^{-6}\,e^{-17.441\,x} \\ +1.177\times10^{-6}\,e^{-26.162\,x} & +5.943\times10^{-7}\,e^{-52.324\,x} \\ +9.234\times10^{-6}\,e^{-56.510\,x} & +2.754\times10^{-6}\,e^{-131.86\,x} \\ -4.673\times10^{-6}\,e^{-261.62\,x} & +1.089\times10^{-5}\,e^{-296.50\,x} \\ -5.656\times10^{-8}\,e^{-313.94\,x} & -7.711\times10^{-6}\,e^{-348.83\,x} \\ +2.464\times10^{-6}\,e^{-523.24\,x} & -1.079\times10^{-6}\,e^{-610.45\,x} \\ +2.868\times10^{-6}\,e^{-627.89\,x} & -2.722\times10^{-6}\,e^{-784.86\,x} \\ -1.686\times10^{-7}\,e^{-872.07\,x} & +1.150\times10^{-6}\,e^{-959.27\,x} \end{pmatrix}}, \quad x = 1-h$$

R6+19 0.479 ppb

$$P = \frac{\pi(a+b)\,R_6(h)}{1-h^{2.2492}\begin{pmatrix} -1.725\times10^{-6}\,e^{-6.2158\,x} & +5.127\times10^{-6}\,e^{-7.9918\,x} \\ -1.950\times10^{-5}\,e^{-13.986\,x} & -6.247\times10^{-7}\,e^{-18.647\,x} \\ +4.278\times10^{-6}\,e^{-21.755\,x} & -2.854\times10^{-6}\,e^{-27.971\,x} \\ +7.848\times10^{-6}\,e^{-55.942\,x} & +9.104\times10^{-6}\,e^{-60.418\,x} \\ -7.917\times10^{-6}\,e^{-74.590\,x} & +6.837\times10^{-6}\,e^{-120.84\,x} \\ +1.165\times10^{-6}\,e^{-140.97\,x} & -1.053\times10^{-5}\,e^{-205.12\,x} \\ +8.362\times10^{-7}\,e^{-242.42\,x} & +1.068\times10^{-5}\,e^{-259.29\,x} \\ +1.017\times10^{-5}\,e^{-279.71\,x} & -1.555\times10^{-5}\,e^{-317.01\,x} \\ +9.247\times10^{-6}\,e^{-559.42\,x} & -8.320\times10^{-6}\,e^{-652.66\,x} \\ +1.741\times10^{-6}\,e^{-839.14\,x} & \end{pmatrix}}, \quad x = 1-h$$

R6+20 0.769 ppb

$$P = \frac{\pi(a+b)\,R_6(h)}{1-h^{2.1787}\begin{pmatrix} -1.912\times10^{-6}\,e^{-6.1961\,x} & +5.652\times10^{-6}\,e^{-7.9664\,x} \\ -2.116\times10^{-5}\,e^{-13.941\,x} & +2.092\times10^{-6}\,e^{-18.588\,x} \\ +5.199\times10^{-6}\,e^{-21.686\,x} & -6.652\times10^{-6}\,e^{-27.882\,x} \\ +1.139\times10^{-5}\,e^{-55.765\,x} & +1.340\times10^{-5}\,e^{-60.226\,x} \\ -1.817\times10^{-5}\,e^{-74.353\,x} & +1.383\times10^{-5}\,e^{-120.45\,x} \\ +7.616\times10^{-6}\,e^{-140.53\,x} & -1.820\times10^{-5}\,e^{-167.29\,x} \\ +2.752\times10^{-6}\,e^{-204.47\,x} & -9.927\times10^{-6}\,e^{-241.65\,x} \\ +1.423\times10^{-5}\,e^{-258.46\,x} & +1.452\times10^{-5}\,e^{-278.82\,x} \\ -1.695\times10^{-5}\,e^{-316.00\,x} & +7.662\times10^{-6}\,e^{-557.65\,x} \\ -6.862\times10^{-6}\,e^{-650.59\,x} & +1.494\times10^{-6}\,e^{-836.47\,x} \end{pmatrix}}, \quad x = 1-h$$

R6+21 0.922 ppb

$$P = \frac{\pi(a{+}b)\,R_6(h)}{1-h^{2.1577}\left(\begin{array}{ll} -1.538{\times}10^{-6}\,e^{-5.8982\,x} & +4.461{\times}10^{-6}\,e^{-7.5834\,x} \\ -1.422{\times}10^{-5}\,e^{-13.271\,x} & -4.676{\times}10^{-6}\,e^{-17.695\,x} \\ +1.718{\times}10^{-6}\,e^{-20.644\,x} & -1.337{\times}10^{-6}\,e^{-26.542\,x} \\ +9.155{\times}10^{-6}\,e^{-53.084\,x} & +6.809{\times}10^{-6}\,e^{-57.330\,x} \\ -7.501{\times}10^{-6}\,e^{-70.778\,x} & +5.939{\times}10^{-6}\,e^{-114.66\,x} \\ +8.324{\times}10^{-6}\,e^{-133.77\,x} & -1.020{\times}10^{-5}\,e^{-159.25\,x} \\ +4.129{\times}10^{-7}\,e^{-194.64\,x} & -1.091{\times}10^{-5}\,e^{-230.03\,x} \\ +1.230{\times}10^{-5}\,e^{-246.04\,x} & +1.064{\times}10^{-5}\,e^{-265.42\,x} \\ -1.074{\times}10^{-5}\,e^{-300.81\,x} & +4.333{\times}10^{-7}\,e^{-389.28\,x} \\ +3.135{\times}10^{-6}\,e^{-530.84\,x} & -3.050{\times}10^{-6}\,e^{-619.31\,x} \\ +8.394{\times}10^{-7}\,e^{-796.26\,x} & \end{array}\right)}, \quad x = 1{-}h$$

R6+22 0.441 ppb

$$P = \frac{\pi(a{+}b)\,R_6(h)}{1-h^{2.2359}\left(\begin{array}{ll} -1.706{\times}10^{-6}\,e^{-6.2243\,x} & +5.079{\times}10^{-6}\,e^{-8.0027\,x} \\ -1.939{\times}10^{-5}\,e^{-14.005\,x} & -1.351{\times}10^{-6}\,e^{-18.673\,x} \\ +6.091{\times}10^{-6}\,e^{-21.785\,x} & -6.273{\times}10^{-6}\,e^{-28.010\,x} \\ +5.028{\times}10^{-6}\,e^{-37.346\,x} & -2.570{\times}10^{-6}\,e^{-56.019\,x} \\ +1.341{\times}10^{-5}\,e^{-60.501\,x} & -2.174{\times}10^{-6}\,e^{-74.692\,x} \\ -1.754{\times}10^{-6}\,e^{-121.00\,x} & +1.201{\times}10^{-5}\,e^{-141.17\,x} \\ -9.670{\times}10^{-6}\,e^{-168.06\,x} & -3.664{\times}10^{-6}\,e^{-205.40\,x} \\ +7.324{\times}10^{-6}\,e^{-224.08\,x} & -1.062{\times}10^{-5}\,e^{-242.75\,x} \\ +1.143{\times}10^{-5}\,e^{-259.64\,x} & +1.200{\times}10^{-5}\,e^{-280.10\,x} \\ -1.621{\times}10^{-5}\,e^{-317.44\,x} & +1.049{\times}10^{-5}\,e^{-560.19\,x} \\ -9.429{\times}10^{-6}\,e^{-653.56\,x} & +1.932{\times}10^{-6}\,e^{-840.29\,x} \end{array}\right)}, \quad x = 1{-}h$$

Total: 55 formulas across 4 tower levels. The complete machine-readable data (all rates, coefficients, ratios, and errors to full precision) is available in `best_results.csv` in the code repository.